Genuine Multipartite Entanglement Analysis — Molecular System

MOLECULE

Benzene (C6H6)
Basis set   : cc-pVDZ
Charge   : 0
Multiplicity : 1
Total e-   : 42
n-body order : 4

GROUND STATE ENERGIES

E0 (Ha) : -230.75355805
E1 (Ha) : -230.57826157
Gap (Ha) : 0.17529648
HF (Ha) : -230.71878874

CONVEX-ROOF METHODS

Methods run   : gcg
Restarts   : 8

C_LB : coherence witness bound
C_expo: unitary opt (exp map)
C_stf : Stiefel manifold (Alg. B)
C_UB : eigendecomposition bound

ACTIVE SPACE CAS(6,6)

Frozen core orbs : 18
Frozen e-   : 36
Active e-   : 6
Active orbs   : 6
Spin-orbitals   : 12
Hilbert dim   : 400

SPIN PROPERTIES

<S2>   : 0.000000
S   : 0.0000
<Sz>   : 0.000000
Multiplicity : 1
Spin state   : Singlet (S=0)

GME CRITERIA

C_GME = sqrt(2 * min_A (1 - Tr(rho_A^2)))

Convex roof (mixed):
  C_GME(rho) = min Sum p_i C_GME(|psi_i>)

Bounds ordering:
  C_LB <= C_GME <= C_UB

DOUBLE OCCUPANCY

<D> total : 2.961506
<D> / site : 0.493584

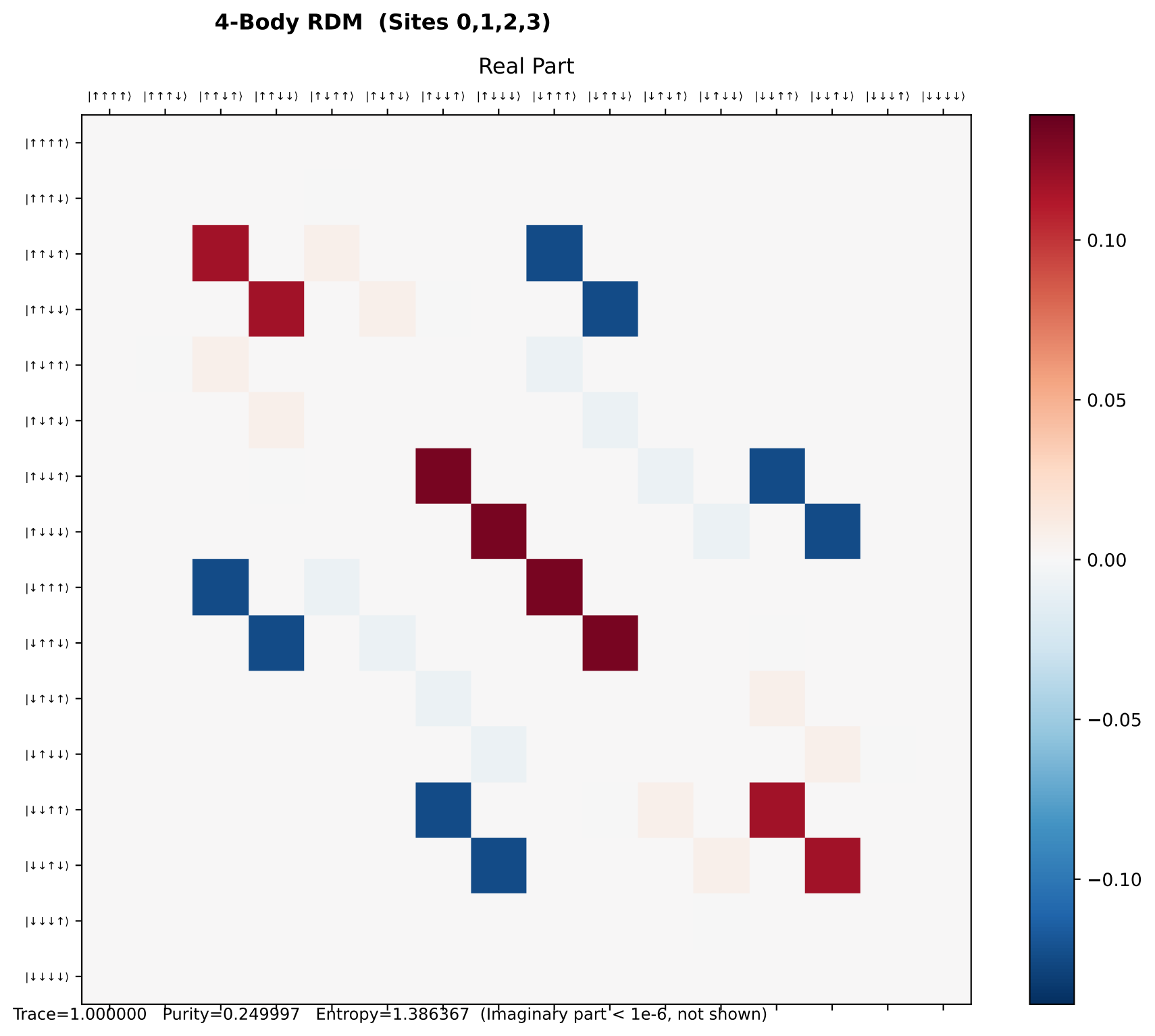
THIS PDF

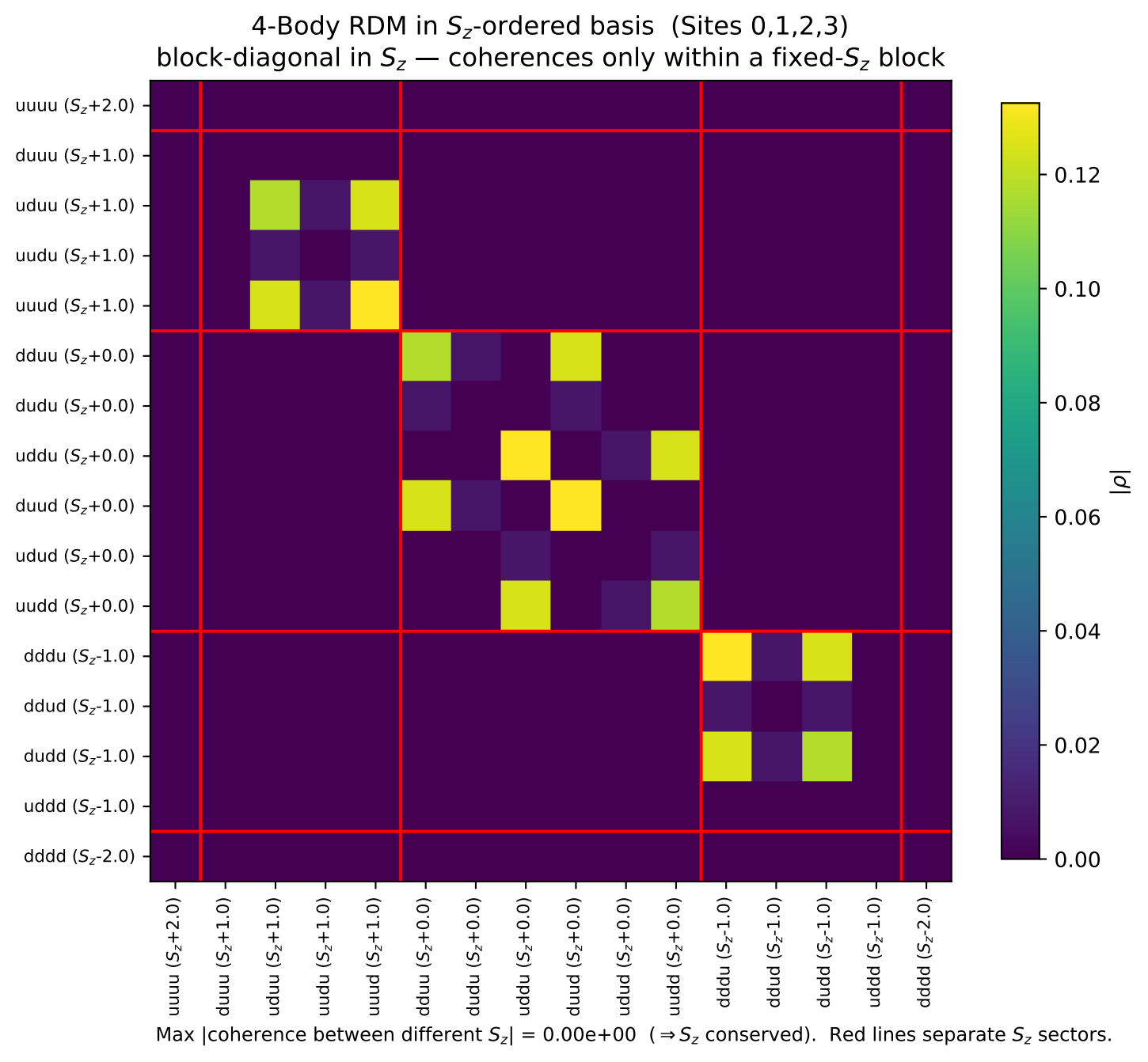
Body order : 4
Clusters   : 15

Date: 2026-07-10 23:30

Summary — 4-Body RDMs (Benzene, CAS(6,6))

Orbitals	Purity	C_GME (pure)	C_LB	C_gcg	C_stiefel	C_UB	State
0,1,2,3	0.249997	—	0.000000	0.000368	—	0.053769	MIXED
0,1,2,4	0.249997	—	0.000000	0.000142	—	0.053767	MIXED
0,1,2,5	0.834591	—	0.471804	0.000000	—	0.090758	WEAKLY MIXED
0,1,3,4	0.249998	—	0.000000	0.000474	—	0.053768	MIXED
0,1,3,5	0.228267	—	0.000000	0.002647	—	0.374537	MIXED
0,1,4,5	0.228210	—	0.000000	0.002539	—	0.375166	MIXED
0,2,3,4	0.249998	—	0.000000	0.000843	—	0.053768	MIXED
0,2,3,5	0.228511	—	0.000000	0.003465	—	0.372263	MIXED
0,2,4,5	0.228456	—	0.000000	0.001915	—	0.372934	MIXED
0,3,4,5	0.801534	—	0.464790	0.000000	—	0.109389	WEAKLY MIXED
1,2,3,4	1.000000	—	0.530061	0.107532	—	0.107532	PURE
1,2,3,5	0.192480	—	0.000000	0.046222	—	0.150506	STRONGLY MIXED
1,2,4,5	0.191842	—	0.000000	0.028805	—	0.153835	STRONGLY MIXED
1,3,4,5	0.212280	—	0.000000	0.044401	—	0.080377	MIXED
2,3,4,5	0.212777	—	0.000000	0.030406	—	0.079753	MIXED


4-Body RDM (Sites 0,1,2,3)
Real Part
|↑↑↑↑⟩ |↑↑↑↓⟩ |↑↑↓↑⟩ |↑↑↓↓⟩ |↑↓↑↑⟩ |↑↓↑↓⟩ |↑↓↓↑⟩ |↑↓↓↓⟩ |↓↑↑↑⟩ |↓↑↑↓⟩ |↓↑↓↑⟩ |↓↑↓↓⟩ |↓↓↑↑⟩ |↓↓↑↓⟩ |↓↓↓↑⟩ |↓↓↓↓⟩
0.10
0.05
0.00
−0.05
−0.10
Trace=1.000000 Purity=0.249997 Entropy=1.386367 (Imaginary part < 1e-6, not shown)


4-Body RDM in S_z-ordered basis (Sites 0,1,2,3)
block-diagonal in S_z — coherences only within a fixed-S_z block
uuuu (S_z+2.0)
duuu (S_z+1.0)
uduu (S_z+1.0)
uudu (S_z+1.0)
uuud (S_z+1.0)
dduu (S_z+0.0)
dudu (S_z+0.0)
uddu (S_z+0.0)
duud (S_z+0.0)
udud (S_z+0.0)
uudd (S_z+0.0)
dddu (S_z-1.0)
ddud (S_z-1.0)
dudd (S_z-1.0)
uddd (S_z-1.0)
dddd (S_z-2.0)
0.12
0.10
0.08
0.06
0.04
0.02
0.00
$|\rho|$
Max |coherence between different S_z| = 0.00e+00 ($\Rightarrow S_z$ conserved). Red lines separate S_z sectors.

Analysis Summary - 4-Body RDM (Sites 0,1,2,3)

RDM Metrics
==========

Trace(rho)	1.0000000000
Purity Tr(rho^2)	0.2499972350
Von Neumann entropy	1.3863667067
Eigenvalue min	-2.760672e-17
Eigenvalue max	2.499991e-01
Imaginary part	< 1e-6 (not shown)

GME Analysis
==========

gamma = Tr(rho^2) = 0.24999724

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.

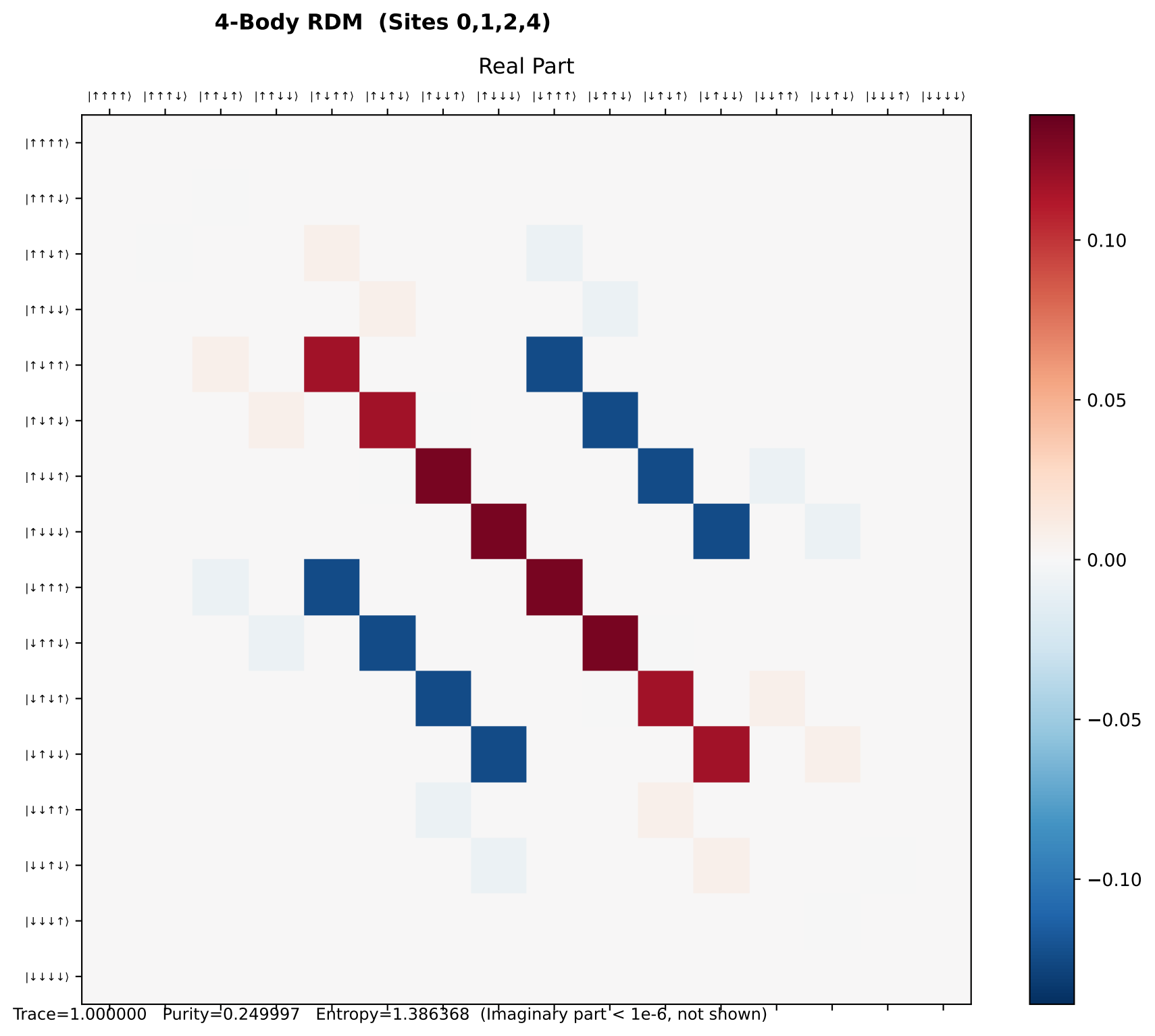
Convex Roof Analysis
==========

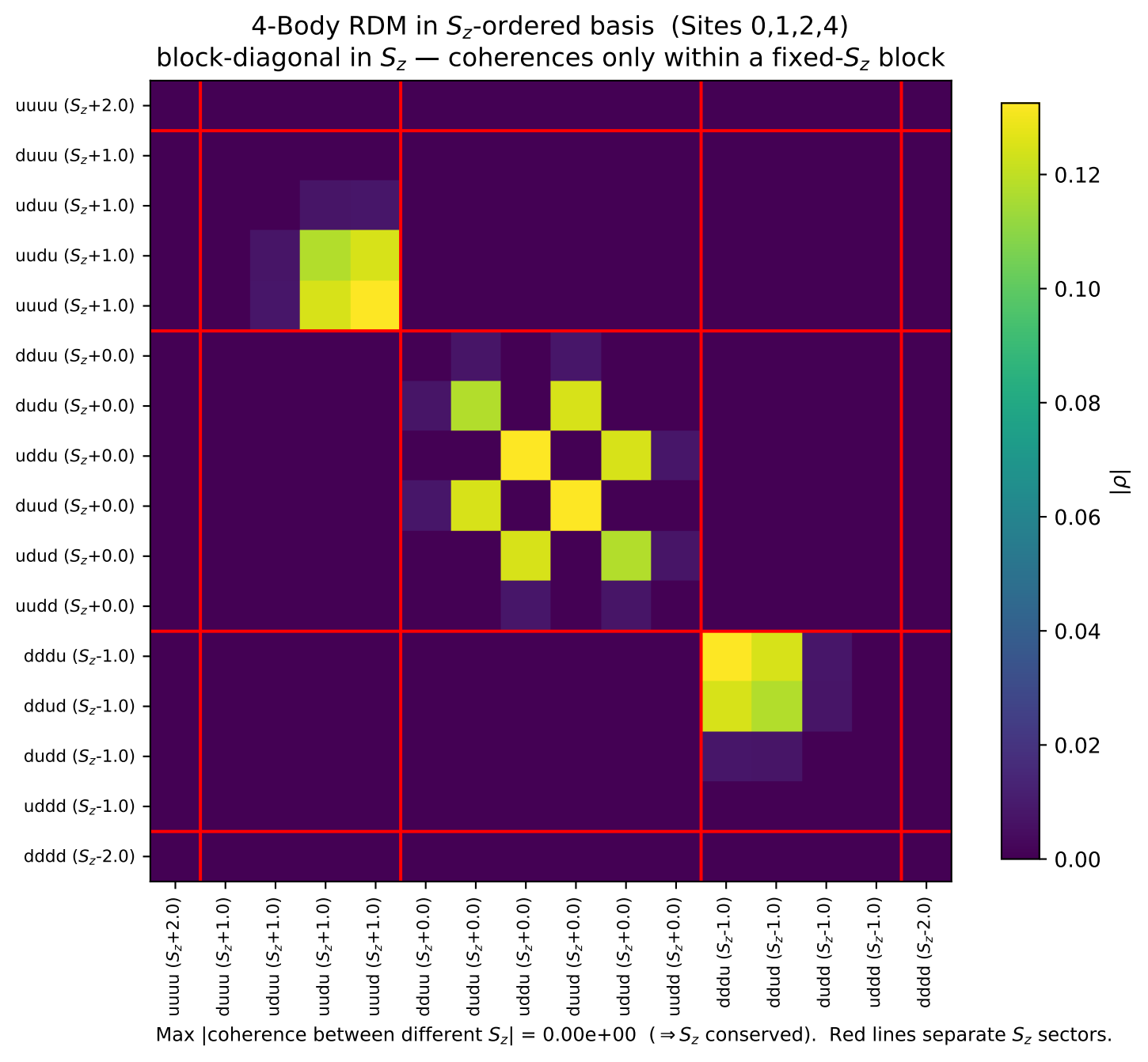
Lower Bounds:
C_LB = 0.00000000
C_GME (unitary optimization):
C_stiefel (unitary opt) = nan BEST
C_GME (unitary optimization - Expoenetial Map):
Upper Bound:
C_UB (eigendecomp) = 0.05376915

Status: inconclusive

Solver source:
–

1. p=0.000139 C_GME=0.151577
2. p=0.000228 C_GME=0.075434
3. p=0.000123 C_GME=0.174821
4. p=0.000038 C_GME=0.225404
5. p=0.000651 C_GME=0.106132
... and 5 more states


4-Body RDM (Sites 0,1,2,4)
Real Part
|↑↑↑↑⟩ |↑↑↑↓⟩ |↑↑↓↑⟩ |↑↑↓↓⟩ |↑↓↑↑⟩ |↑↓↑↓⟩ |↑↓↓↑⟩ |↑↓↓↓⟩ |↓↑↑↑⟩ |↓↑↑↓⟩ |↓↑↓↑⟩ |↓↑↓↓⟩ |↓↓↑↑⟩ |↓↓↑↓⟩ |↓↓↓↑⟩ |↓↓↓↓⟩
0.10
0.05
0.00
−0.05
−0.10
Trace=1.000000 Purity=0.249997 Entropy=1.386368 (Imaginary part < 1e-6, not shown)


4-Body RDM in S_z-ordered basis (Sites 0,1,2,4)
block-diagonal in S_z — coherences only within a fixed-S_z block
uuuu (S_z+2.0)
duuu (S_z+1.0)
uduu (S_z+1.0)
uudu (S_z+1.0)
uuud (S_z+1.0)
dduu (S_z+0.0)
dudu (S_z+0.0)
uddu (S_z+0.0)
duud (S_z+0.0)
udud (S_z+0.0)
uudd (S_z+0.0)
dddu (S_z-1.0)
ddud (S_z-1.0)
dudd (S_z-1.0)
uddd (S_z-1.0)
dddd (S_z-2.0)
0.12
0.10
0.08
0.06
0.04
0.02
0.00
$|\rho|$
Max |coherence between different S_z| = 0.00e+00 ($\Rightarrow S_z$ conserved). Red lines separate S_z sectors.

Analysis Summary - 4-Body RDM (Sites 0,1,2,4)

RDM Metrics
==========

Trace(rho)	1.0000000000
Purity Tr(rho^2)	0.2499971871
Von Neumann entropy	1.3863678625
Eigenvalue min	-1.905566e-17
Eigenvalue max	2.499991e-01
Imaginary part	< 1e-6 (not shown)

GME Analysis
==========

gamma = Tr(rho^2) = 0.24999719

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.

Convex Roof Analysis
==========

Lower Bounds:
- C_LB = 0.00000000

C_GME (unitary optimization):
- C_stiefel (unitary opt) = nan BEST

C_GME (unitary optimization - Expoenetial Map):

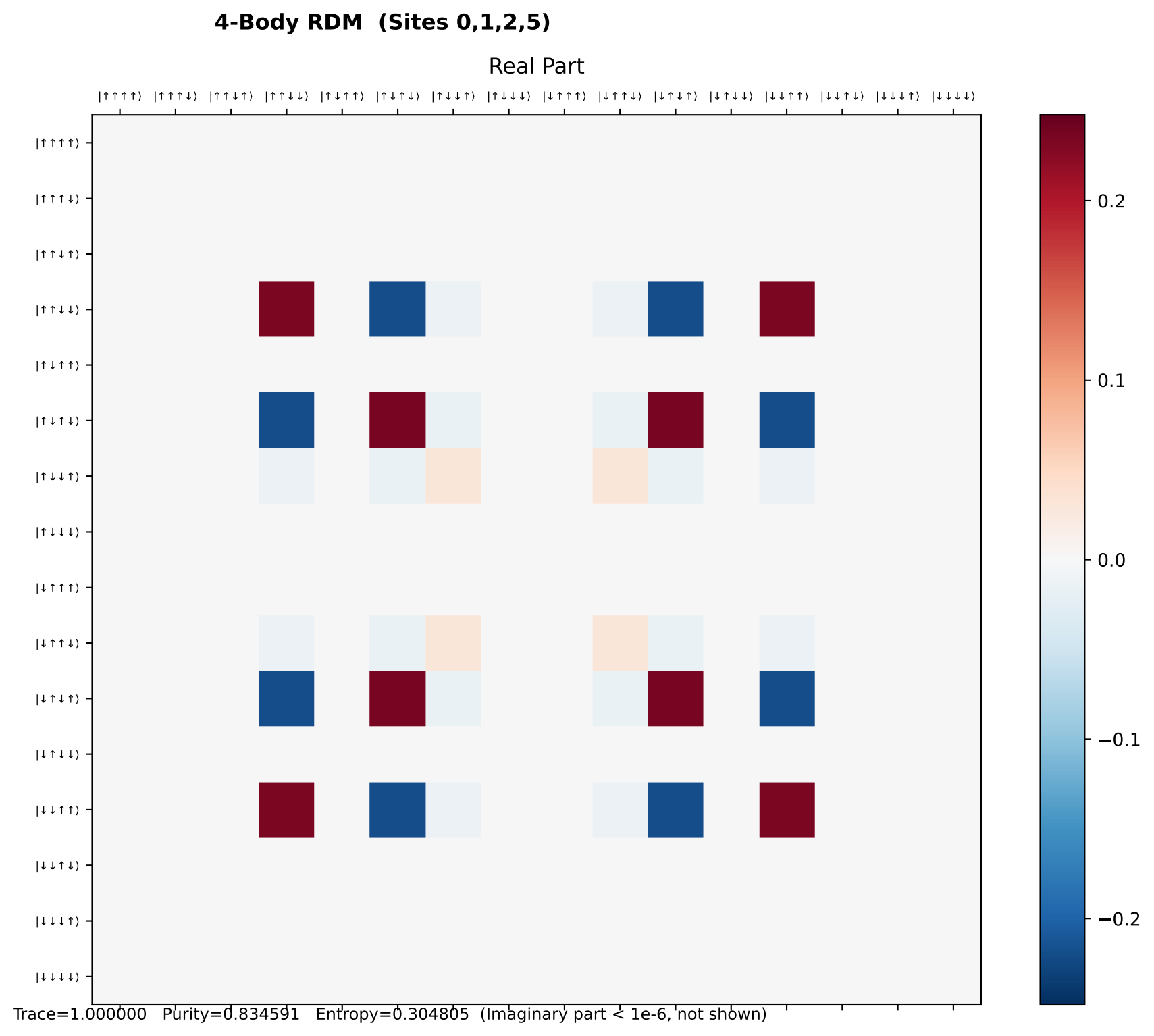
Upper Bound:
- C_UB (eigendecomp) = 0.05376745

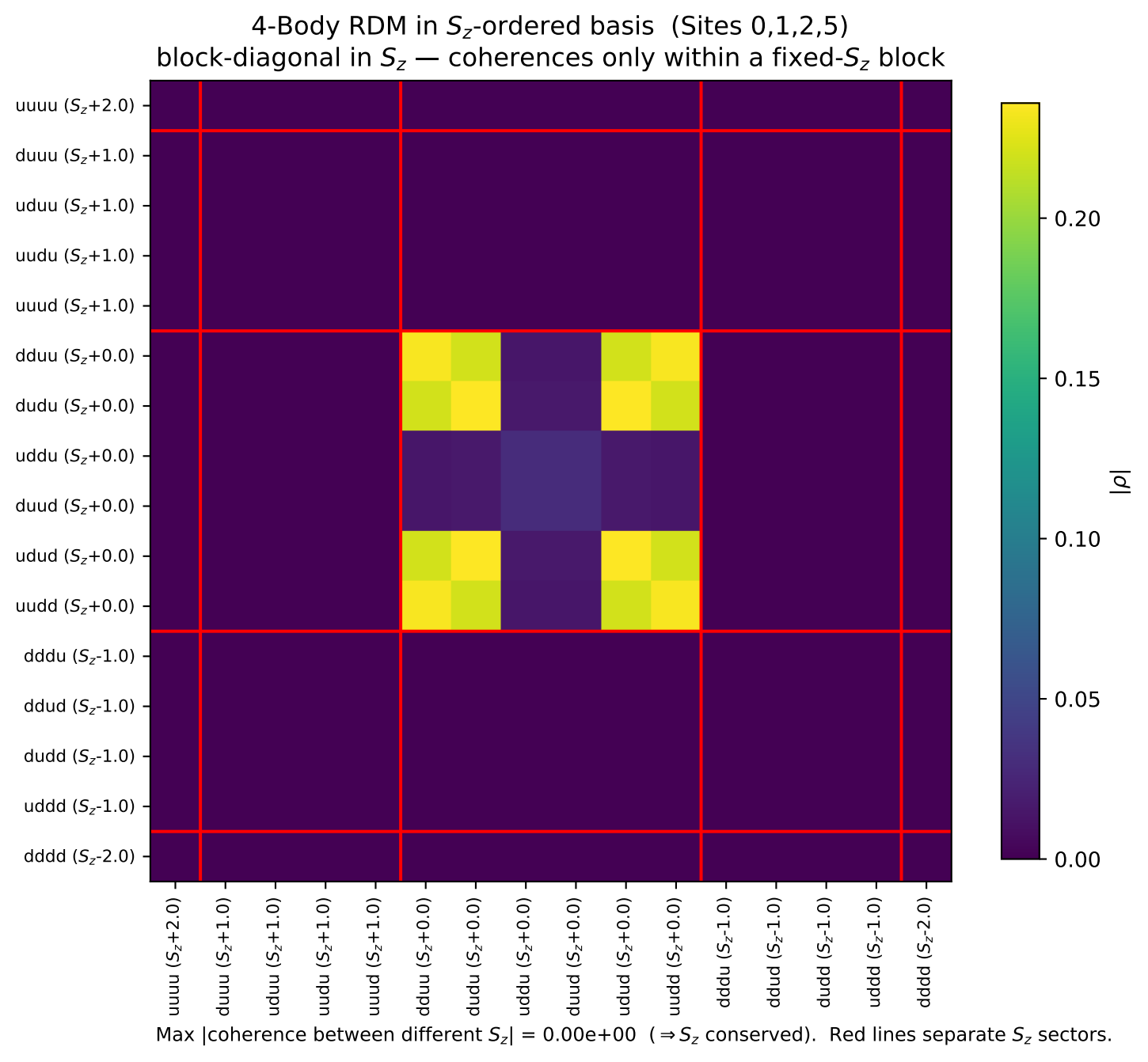
Status: inconclusive

Solver source:
- –

1. p=0.249820 C_GME=0.000056
2. p=0.249831 C_GME=0.000036
3. p=0.000027 C_GME=0.204944
4. p=0.000235 C_GME=0.080509
5. p=0.000075 C_GME=0.076969

... and 5 more states


4-Body RDM (Sites 0,1,2,5)
Real Part
|↑↑↑↑⟩ |↑↑↑↓⟩ |↑↑↓↑⟩ |↑↑↓↓⟩ |↑↓↑↑⟩ |↑↓↑↓⟩ |↑↓↓↑⟩ |↑↓↓↓⟩ |↓↑↑↑⟩ |↓↑↑↓⟩ |↓↑↓↑⟩ |↓↑↓↓⟩ |↓↓↑↑⟩ |↓↓↑↓⟩ |↓↓↓↑⟩ |↓↓↓↓⟩
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=0.834591 Entropy=0.304805 (Imaginary part < 1e-6, not shown)


4-Body RDM in S_z-ordered basis (Sites 0,1,2,5)
block-diagonal in S_z — coherences only within a fixed-S_z block
uuuu (S_z+2.0)
duuu (S_z+1.0)
uduu (S_z+1.0)
uudu (S_z+1.0)
uuud (S_z+1.0)
dduu (S_z+0.0)
dudu (S_z+0.0)
uddu (S_z+0.0)
duud (S_z+0.0)
udud (S_z+0.0)
uudd (S_z+0.0)
dddu (S_z-1.0)
ddud (S_z-1.0)
dudd (S_z-1.0)
uddd (S_z-1.0)
dddd (S_z-2.0)
0.20
0.15
0.10
0.05
0.00
$|\rho|$
Max |coherence between different S_z| = 0.00e+00 ($\Rightarrow S_z$ conserved). Red lines separate S_z sectors.

Analysis Summary - 4-Body RDM (Sites 0,1,2,5)

RDM Metrics
==========

Trace(rho)	1.0000000000
Purity Tr(rho^2)	0.8345909728
Von Neumann entropy	0.3048046140
Eigenvalue min	-7.196502e-17
Eigenvalue max	9.090177e-01
Imaginary part	< 1e-6 (not shown)

GME Analysis
==========

gamma = Tr(rho^2) = 0.83459097

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.

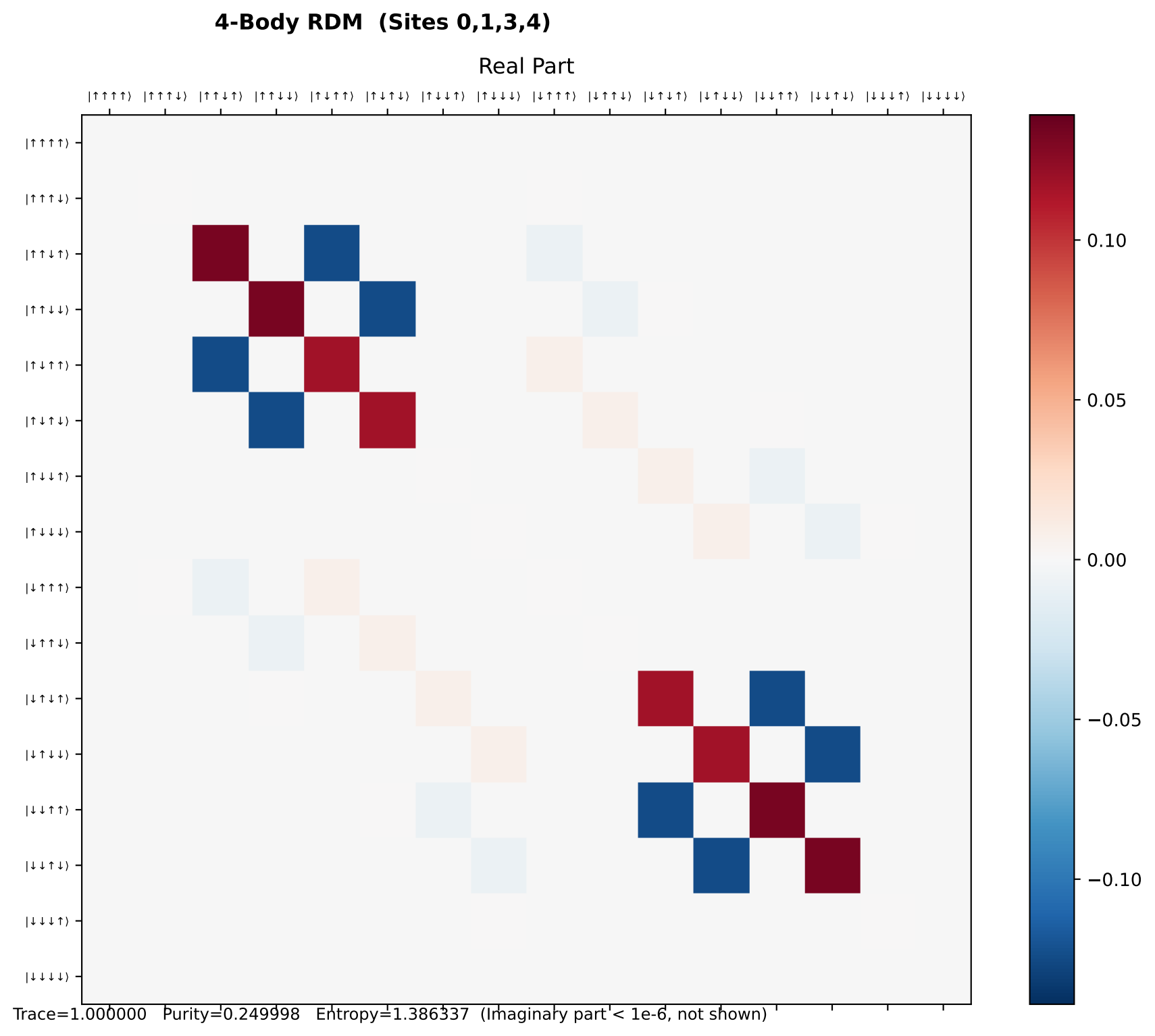
Convex Roof Analysis
==========

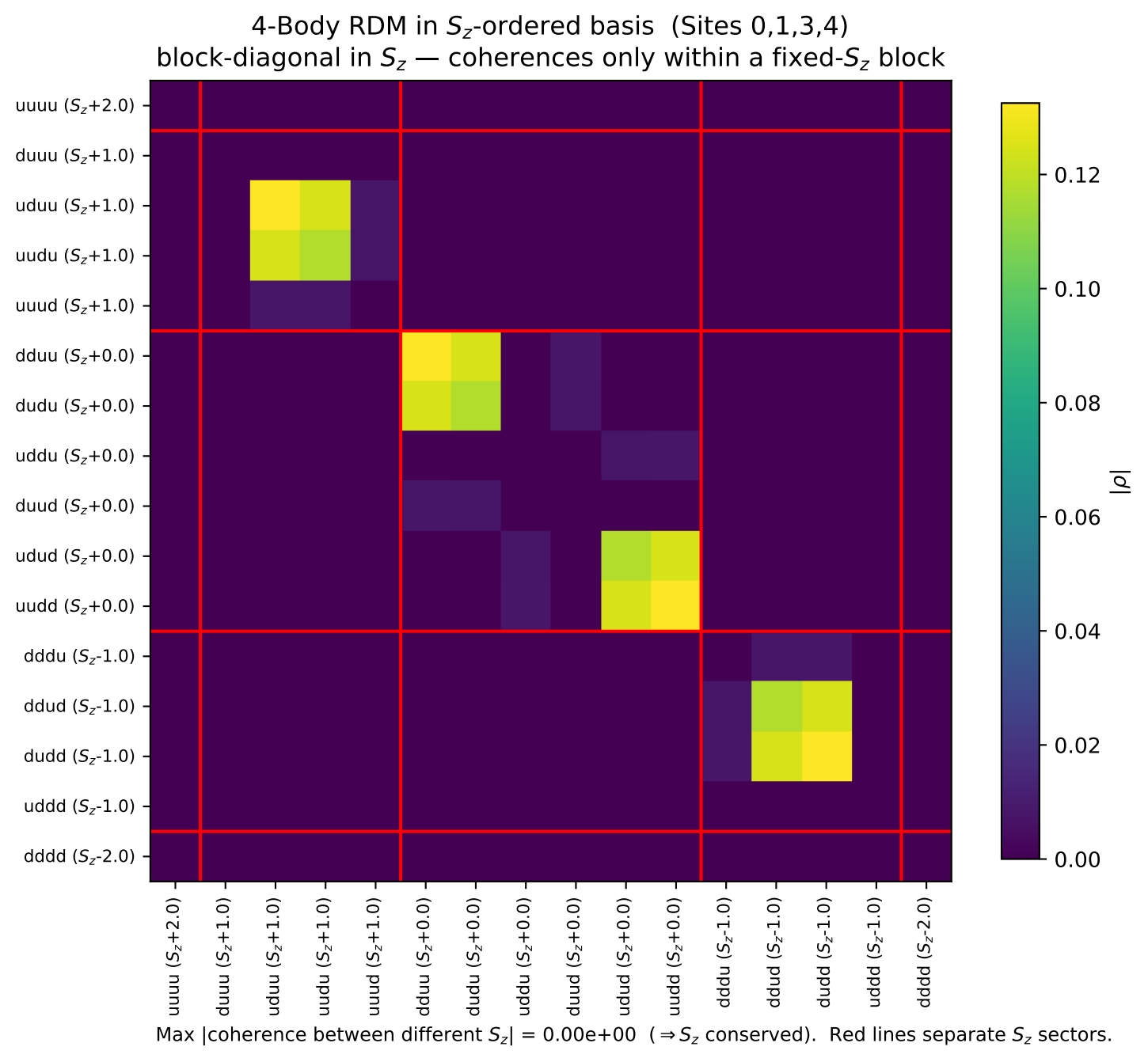
Lower Bounds:
C_LB = 0.47180392
C_GME (unitary optimization):
C_stiefel (unitary opt) = nan BEST
C_GME (unitary optimization - Expoenetial Map):
Upper Bound:
C_UB (eigendecomp) = 0.09075800

Status: no_GME

Solver source:
–

1. p=0.535704 C_GME=0.000000
2. p=0.342963 C_GME=0.000000
3. p=0.064940 C_GME=0.000000
4. p=0.056392 C_GME=0.000002


4-Body RDM (Sites 0,1,3,4)
Real Part
|↑↑↑↑⟩ |↑↑↑↓⟩ |↑↑↓↑⟩ |↑↑↓↓⟩ |↑↓↑↑⟩ |↑↓↑↓⟩ |↑↓↓↑⟩ |↑↓↓↓⟩ |↓↑↑↑⟩ |↓↑↑↓⟩ |↓↑↓↑⟩ |↓↑↓↓⟩ |↓↓↑↑⟩ |↓↓↑↓⟩ |↓↓↓↑⟩ |↓↓↓↓⟩
0.10
0.05
0.00
−0.05
−0.10
Trace=1.000000 Purity=0.249998 Entropy=1.386337 (Imaginary part < 1e-6, not shown)


4-Body RDM in S_z-ordered basis (Sites 0,1,3,4)
block-diagonal in S_z — coherences only within a fixed-S_z block
uuuu (S_z+2.0)
duuu (S_z+1.0)
uduu (S_z+1.0)
uudu (S_z+1.0)
uuud (S_z+1.0)
dduu (S_z+0.0)
dudu (S_z+0.0)
uddu (S_z+0.0)
duud (S_z+0.0)
udud (S_z+0.0)
uudd (S_z+0.0)
dddu (S_z-1.0)
ddud (S_z-1.0)
dudd (S_z-1.0)
uddd (S_z-1.0)
dddd (S_z-2.0)
0.12
0.10
0.08
0.06
0.04
0.02
0.00
$|\rho|$
Max |coherence between different S_z| = 0.00e+00 ($\Rightarrow S_z$ conserved). Red lines separate S_z sectors.

Analysis Summary - 4-Body RDM (Sites 0,1,3,4)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.2499984586
  Von Neumann entropy  : 1.3863365325
  Eigenvalue min       : -7.644474e-18
  Eigenvalue max       : 2.499993e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 0.24999846

  A            | B            | Tr(rho_A^2)
  ─────────────| ─────────────| ──────────

STATUS: MIXED STATE
──────────────────────────────────

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

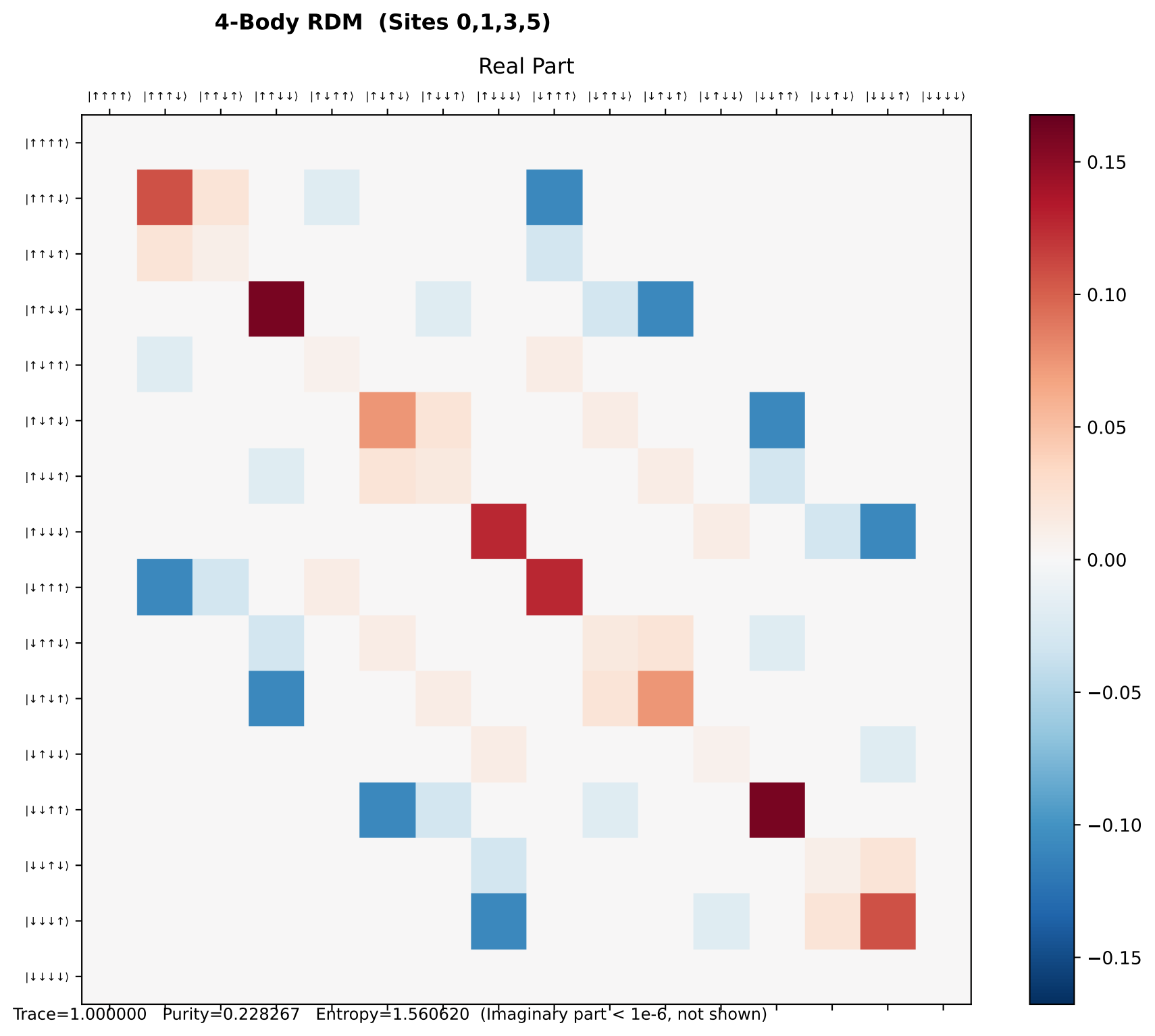
```
Convex Roof Analysis
====================================

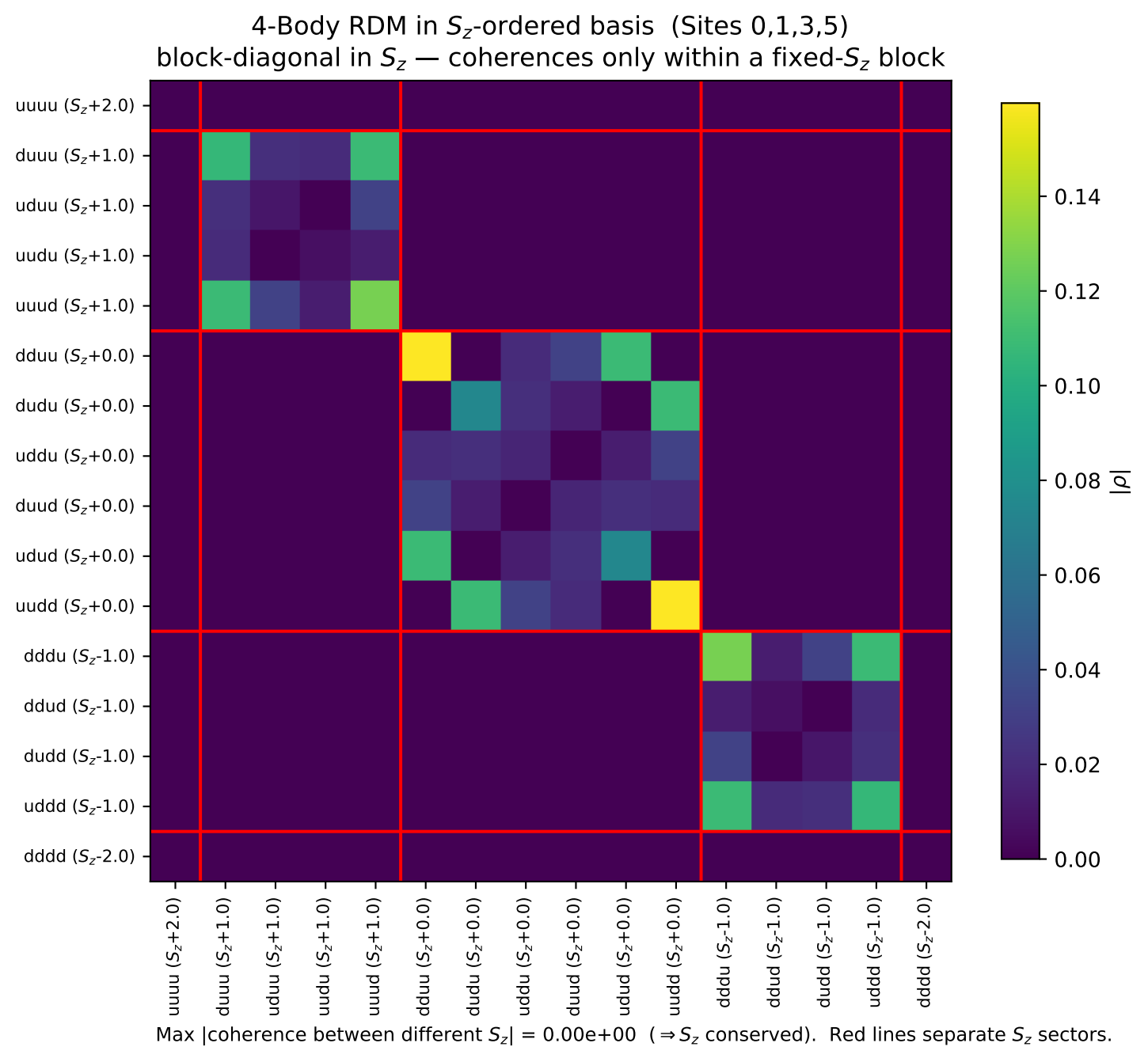
  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = nan   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.05376780

  Status: inconclusive

  Solver source:
    —
```

```
1. p=0.000869  C_GME=0.084245
2. p=0.248744  C_GME=0.000106
3. p=0.247773  C_GME=0.000238
4. p=0.000873  C_GME=0.026764
5. p=0.001630  C_GME=0.076547
... and 5 more states
```


4-Body RDM (Sites 0,1,3,5)
Real Part
|↑↑↑↑⟩ |↑↑↑↓⟩ |↑↑↓↑⟩ |↑↑↓↓⟩ |↑↓↑↑⟩ |↑↓↑↓⟩ |↑↓↓↑⟩ |↑↓↓↓⟩ |↓↑↑↑⟩ |↓↑↑↓⟩ |↓↑↓↑⟩ |↓↑↓↓⟩ |↓↓↑↑⟩ |↓↓↑↓⟩ |↓↓↓↑⟩ |↓↓↓↓⟩
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
Trace=1.000000 Purity=0.228267 Entropy=1.560620 (Imaginary part < 1e-6, not shown)


4-Body RDM in S_z-ordered basis (Sites 0,1,3,5)
block-diagonal in S_z — coherences only within a fixed-S_z block
uuuu (S_z+2.0)
duuu (S_z+1.0)
uduu (S_z+1.0)
uudu (S_z+1.0)
uuud (S_z+1.0)
dduu (S_z+0.0)
dudu (S_z+0.0)
uddu (S_z+0.0)
duud (S_z+0.0)
udud (S_z+0.0)
uudd (S_z+0.0)
dddu (S_z-1.0)
ddud (S_z-1.0)
dudd (S_z-1.0)
uddd (S_z-1.0)
dddd (S_z-2.0)
0.14
0.12
0.10
0.08
0.06
0.04
0.02
0.00
$|\rho|$
Max |coherence between different S_z| = 0.00e+00 ($\Rightarrow S_z$ conserved). Red lines separate S_z sectors.

Analysis Summary - 4-Body RDM (Sites 0,1,3,5)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.2282670771
  Von Neumann entropy  : 1.5606201050
  Eigenvalue min       : -3.947696e-17
  Eigenvalue max       : 2.499215e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 0.22826708

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

STATUS: MIXED STATE
-------------------

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

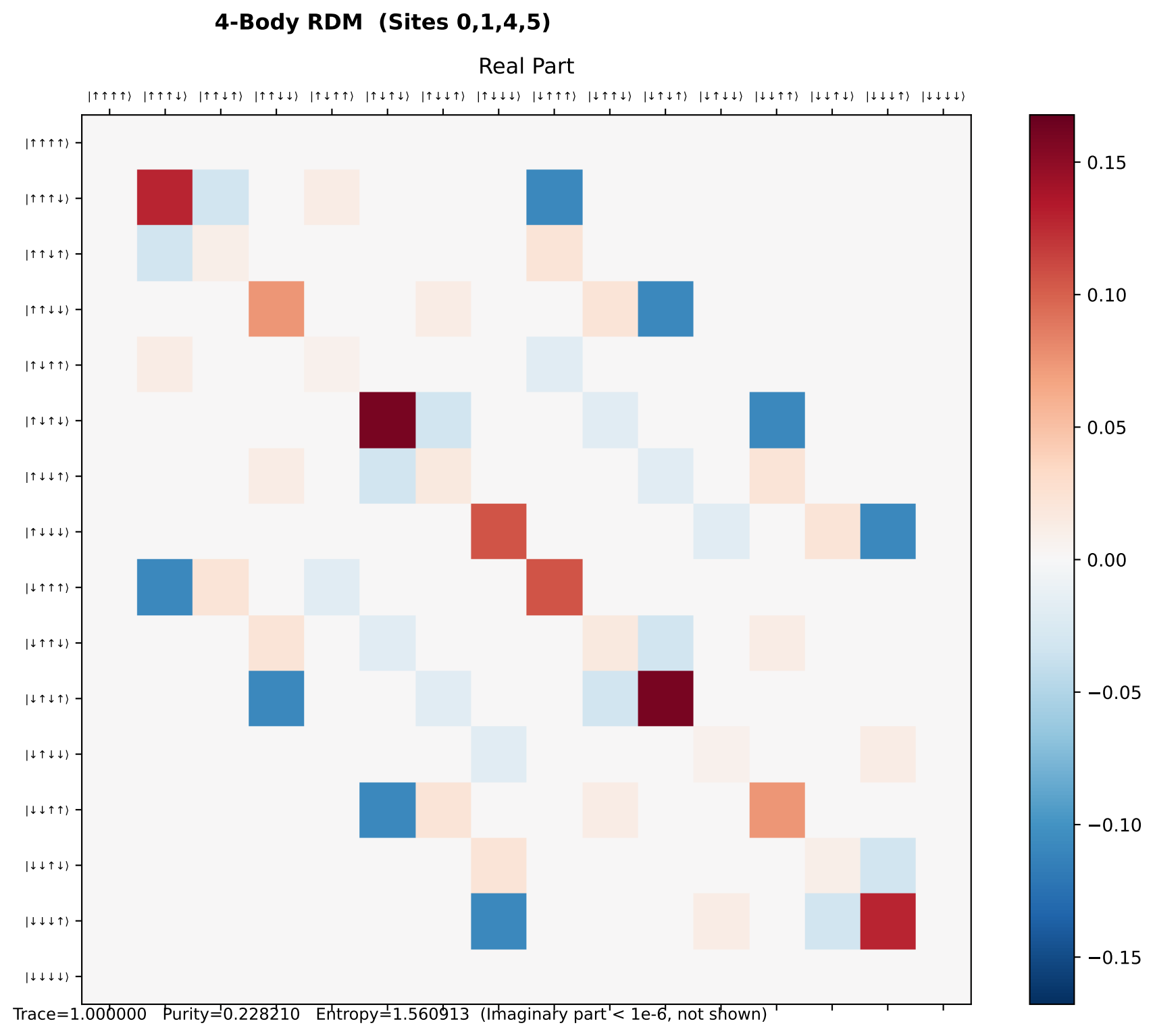
```
Convex Roof Analysis
====================================

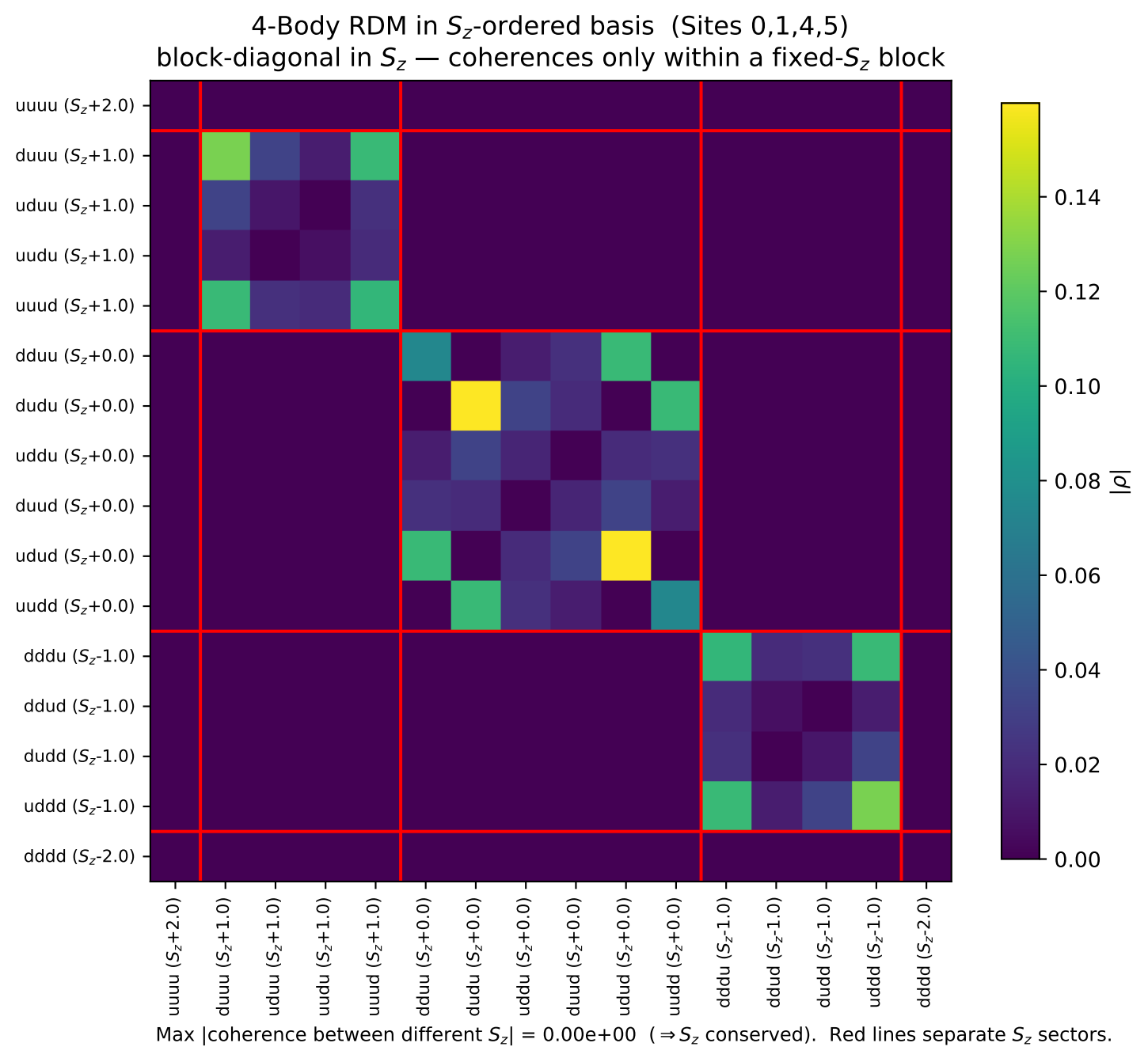
  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = nan   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.37453748

  Status: inconclusive

  Solver source:
    –
```

```
1. p=0.164269  C_GME=0.001594
2. p=0.022555  C_GME=0.016568
3. p=0.082594  C_GME=0.001666
4. p=0.085586  C_GME=0.002630
5. p=0.025236  C_GME=0.012541
... and 5 more states
```


4-Body RDM (Sites 0,1,4,5)
Real Part
|↑↑↑↑⟩ |↑↑↑↓⟩ |↑↑↓↑⟩ |↑↑↓↓⟩ |↑↓↑↑⟩ |↑↓↑↓⟩ |↑↓↓↑⟩ |↑↓↓↓⟩ |↓↑↑↑⟩ |↓↑↑↓⟩ |↓↑↓↑⟩ |↓↑↓↓⟩ |↓↓↑↑⟩ |↓↓↑↓⟩ |↓↓↓↑⟩ |↓↓↓↓⟩
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
Trace=1.000000 Purity=0.228210 Entropy=1.560913 (Imaginary part < 1e-6, not shown)


4-Body RDM in S_z-ordered basis (Sites 0,1,4,5)
block-diagonal in S_z — coherences only within a fixed-S_z block
uuuu (S_z+2.0)
duuu (S_z+1.0)
uduu (S_z+1.0)
uudu (S_z+1.0)
uuud (S_z+1.0)
dduu (S_z+0.0)
dudu (S_z+0.0)
uddu (S_z+0.0)
duud (S_z+0.0)
udud (S_z+0.0)
uudd (S_z+0.0)
dddu (S_z-1.0)
ddud (S_z-1.0)
dudd (S_z-1.0)
uddd (S_z-1.0)
dddd (S_z-2.0)
0.14
0.12
0.10
0.08
0.06
0.04
0.02
0.00
$|\rho|$
Max |coherence between different S_z| = 0.00e+00 ($\Rightarrow S_z$ conserved). Red lines separate S_z sectors.

Analysis Summary - 4-Body RDM (Sites 0,1,4,5)

RDM Metrics
==========

Trace(rho)	1.0000000000
Purity Tr(rho^2)	0.2282097687
Von Neumann entropy	1.5609134306
Eigenvalue min	-3.331269e-17
Eigenvalue max	2.499339e-01
Imaginary part	< 1e-6 (not shown)

GME Analysis
==========

gamma = Tr(rho^2) = 0.22820977

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.

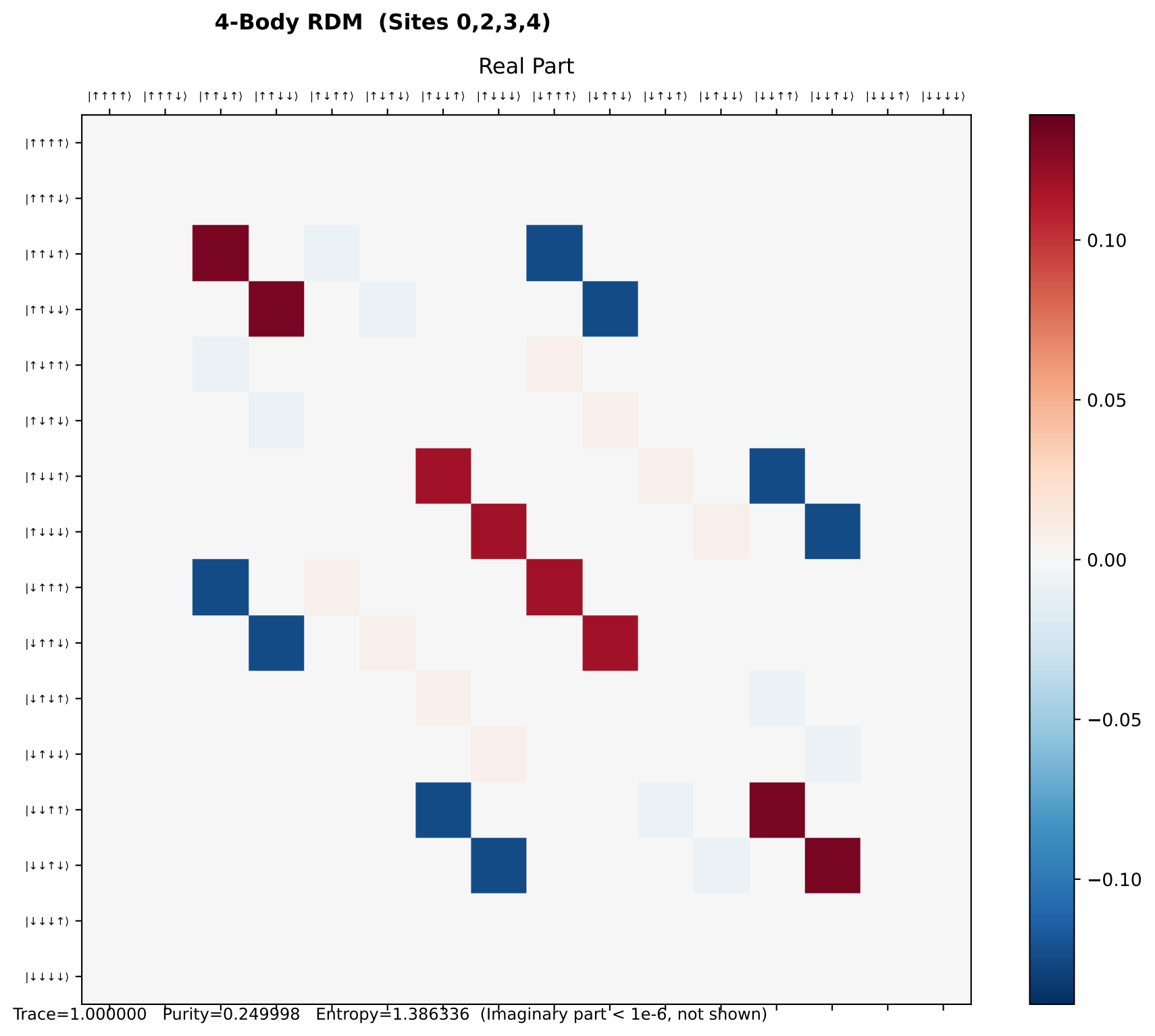
Convex Roof Analysis
==========

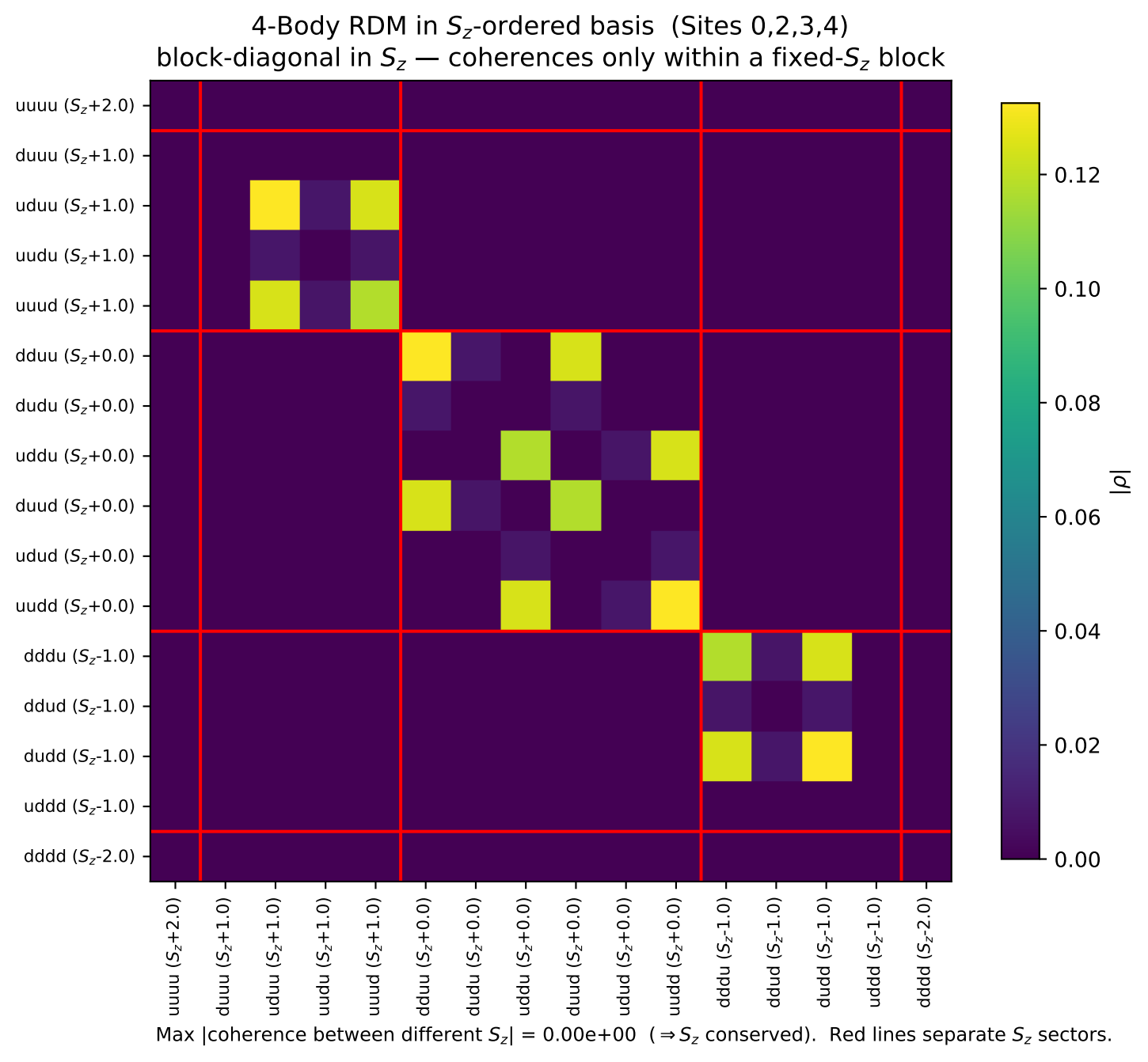
Lower Bounds:
C_LB = 0.00000000
C_GME (unitary optimization):
C_stiefel (unitary opt) = nan BEST
C_GME (unitary optimization - Expoenetial Map):
Upper Bound:
C_UB (eigendecomp) = 0.37516586

Status: inconclusive

Solver source:
–

1. p=0.071815 C_GME=0.004945
2. p=0.161250 C_GME=0.000917
3. p=0.023830 C_GME=0.020212
4. p=0.084099 C_GME=0.003667
5. p=0.088153 C_GME=0.002998
... and 5 more states


4-Body RDM (Sites 0,2,3,4)
Real Part
|↑↑↑↑⟩ |↑↑↑↓⟩ |↑↑↓↑⟩ |↑↑↓↓⟩ |↑↓↑↑⟩ |↑↓↑↓⟩ |↑↓↓↑⟩ |↑↓↓↓⟩ |↓↑↑↑⟩ |↓↑↑↓⟩ |↓↑↓↑⟩ |↓↑↓↓⟩ |↓↓↑↑⟩ |↓↓↑↓⟩ |↓↓↓↑⟩ |↓↓↓↓⟩
0.10
0.05
0.00
−0.05
−0.10
Trace=1.000000 Purity=0.249998 Entropy=1.386336 (Imaginary part < 1e-6, not shown)


4-Body RDM in S_z-ordered basis (Sites 0,2,3,4)
block-diagonal in S_z — coherences only within a fixed-S_z block
uuuu (S_z+2.0)
duuu (S_z+1.0)
uduu (S_z+1.0)
uudu (S_z+1.0)
uuud (S_z+1.0)
dduu (S_z+0.0)
dudu (S_z+0.0)
uddu (S_z+0.0)
duud (S_z+0.0)
udud (S_z+0.0)
uudd (S_z+0.0)
dddu (S_z-1.0)
ddud (S_z-1.0)
dudd (S_z-1.0)
uddd (S_z-1.0)
dddd (S_z-2.0)
$|\rho|$
0.12
0.10
0.08
0.06
0.04
0.02
0.00
Max |coherence between different S_z| = 0.00e+00 ($\Rightarrow S_z$ conserved). Red lines separate S_z sectors.

Analysis Summary - 4-Body RDM (Sites 0,2,3,4)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.2499984844
  Von Neumann entropy  : 1.3863358790
  Eigenvalue min       : -5.359985e-17
  Eigenvalue max       : 2.499993e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 0.24999848

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

STATUS: MIXED STATE
-------------------

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

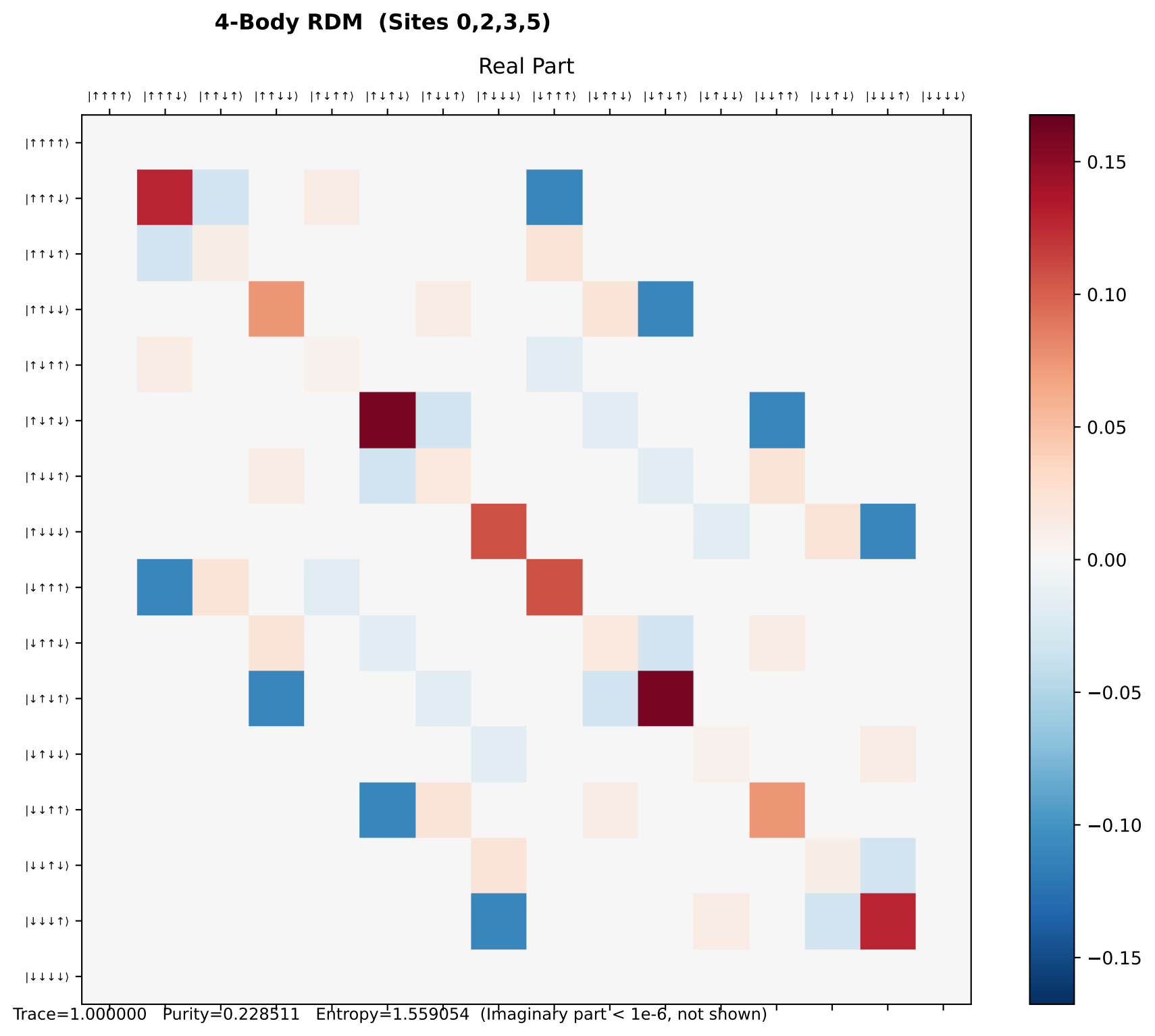
```
Convex Roof Analysis
====================================

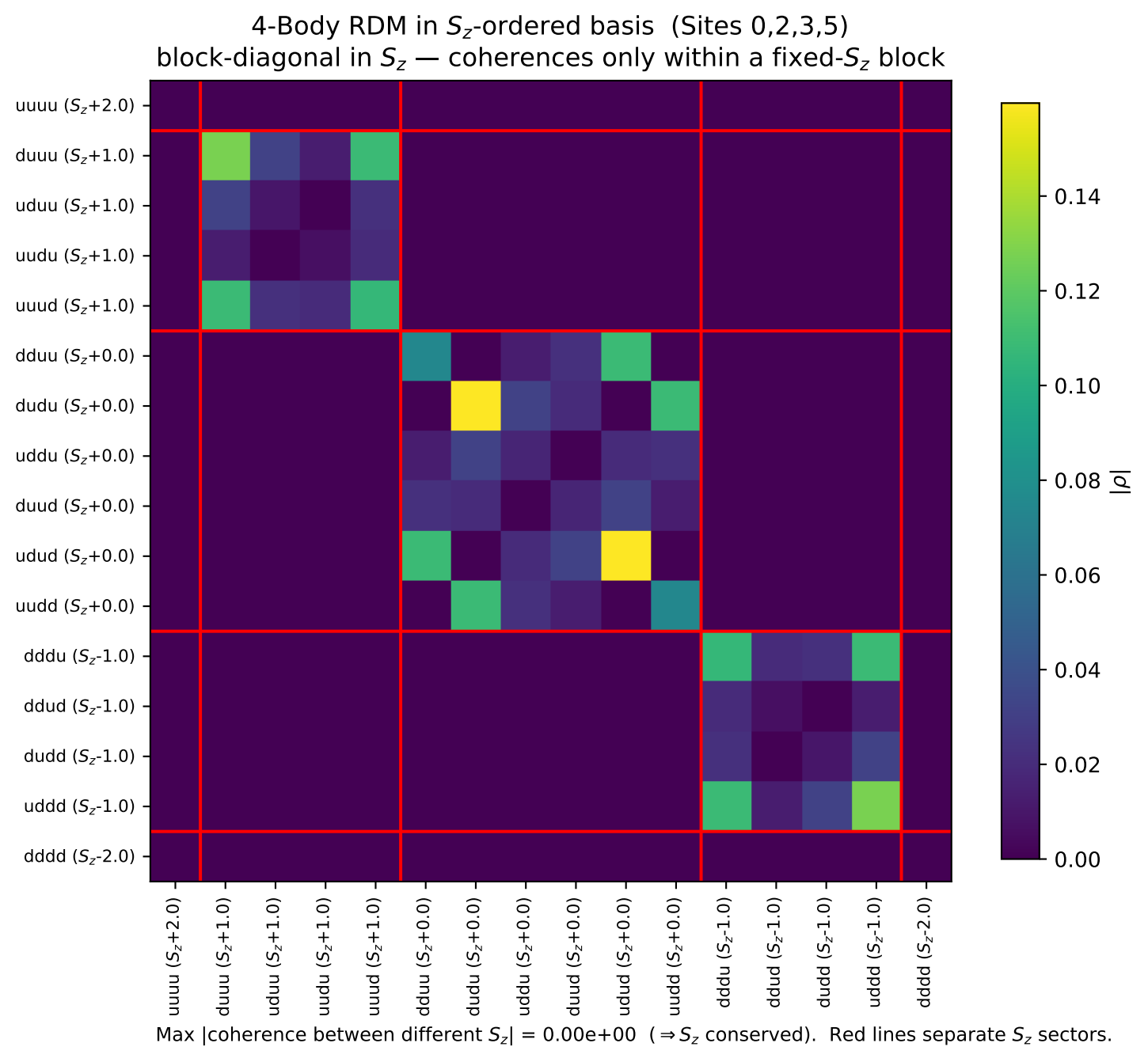
  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = nan   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.05376776

  Status: inconclusive

  Solver source:
    –
```

```
1. p=0.249692  C_GME=0.000064
2. p=0.133356  C_GME=0.000929
3. p=0.247620  C_GME=0.000327
4. p=0.000983  C_GME=0.022378
5. p=0.011491  C_GME=0.010993
... and 5 more states
```


4-Body RDM (Sites 0,2,3,5)
Real Part
|↑↑↑↑⟩ |↑↑↑↓⟩ |↑↑↓↑⟩ |↑↑↓↓⟩ |↑↓↑↑⟩ |↑↓↑↓⟩ |↑↓↓↑⟩ |↑↓↓↓⟩ |↓↑↑↑⟩ |↓↑↑↓⟩ |↓↑↓↑⟩ |↓↑↓↓⟩ |↓↓↑↑⟩ |↓↓↑↓⟩ |↓↓↓↑⟩ |↓↓↓↓⟩
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
Trace=1.000000 Purity=0.228511 Entropy=1.559054 (Imaginary part < 1e-6, not shown)


4-Body RDM in S_z-ordered basis (Sites 0,2,3,5)
block-diagonal in S_z — coherences only within a fixed-S_z block
uuuu (S_z+2.0)
duuu (S_z+1.0)
uduu (S_z+1.0)
uudu (S_z+1.0)
uuud (S_z+1.0)
dduu (S_z+0.0)
dudu (S_z+0.0)
uddu (S_z+0.0)
duud (S_z+0.0)
udud (S_z+0.0)
uudd (S_z+0.0)
dddu (S_z-1.0)
ddud (S_z-1.0)
dudd (S_z-1.0)
uddd (S_z-1.0)
dddd (S_z-2.0)
0.14
0.12
0.10
0.08
0.06
0.04
0.02
0.00
$|\rho|$
Max |coherence between different S_z| = 0.00e+00 ($\Rightarrow S_z$ conserved). Red lines separate S_z sectors.

Analysis Summary - 4-Body RDM (Sites 0,2,3,5)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.2285113709
  Von Neumann entropy  : 1.5590535668
  Eigenvalue min       : -5.409989e-17
  Eigenvalue max       : 2.499308e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 0.22851137

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

STATUS: MIXED STATE
-------------------

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

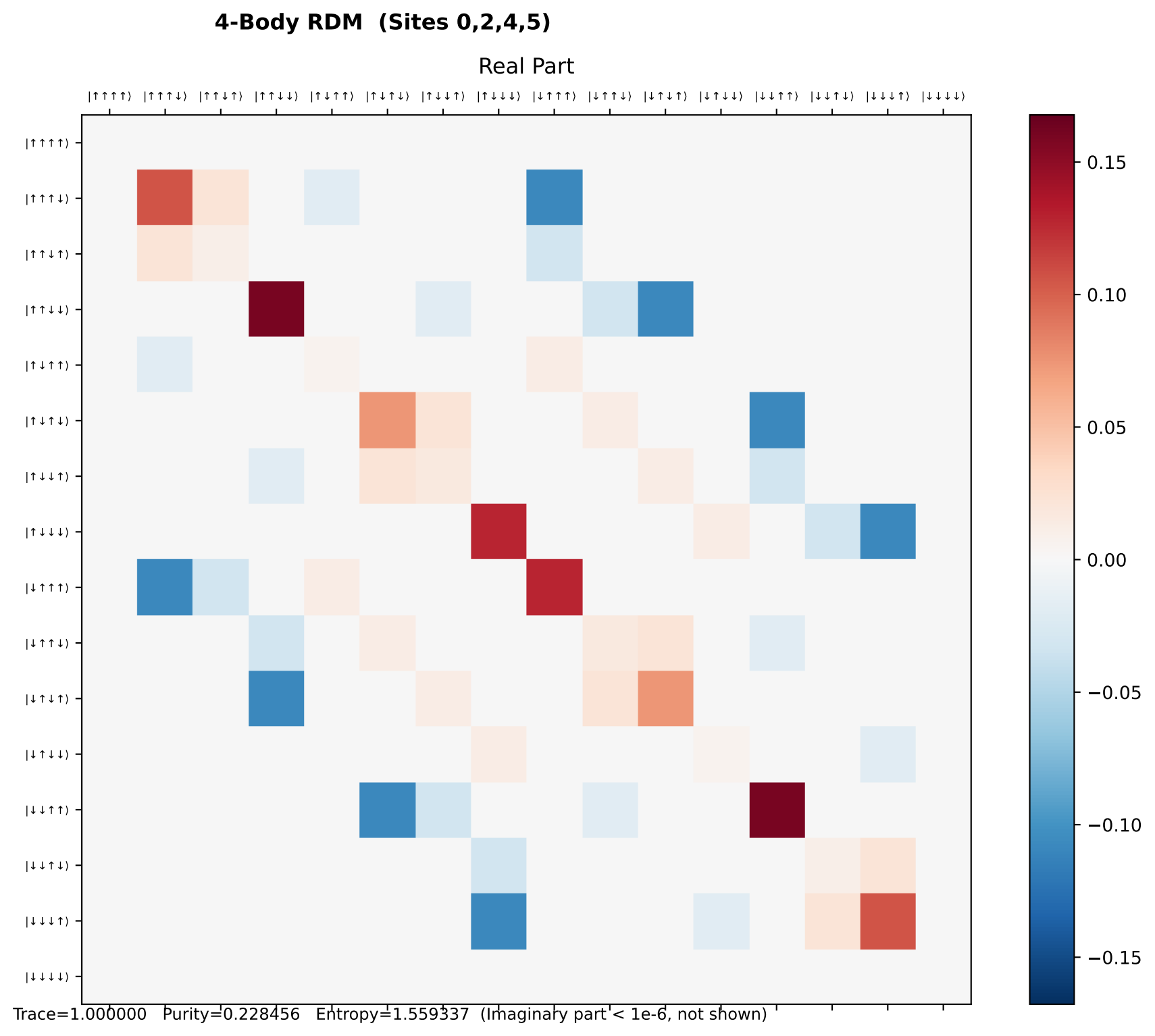
```
Convex Roof Analysis
====================================

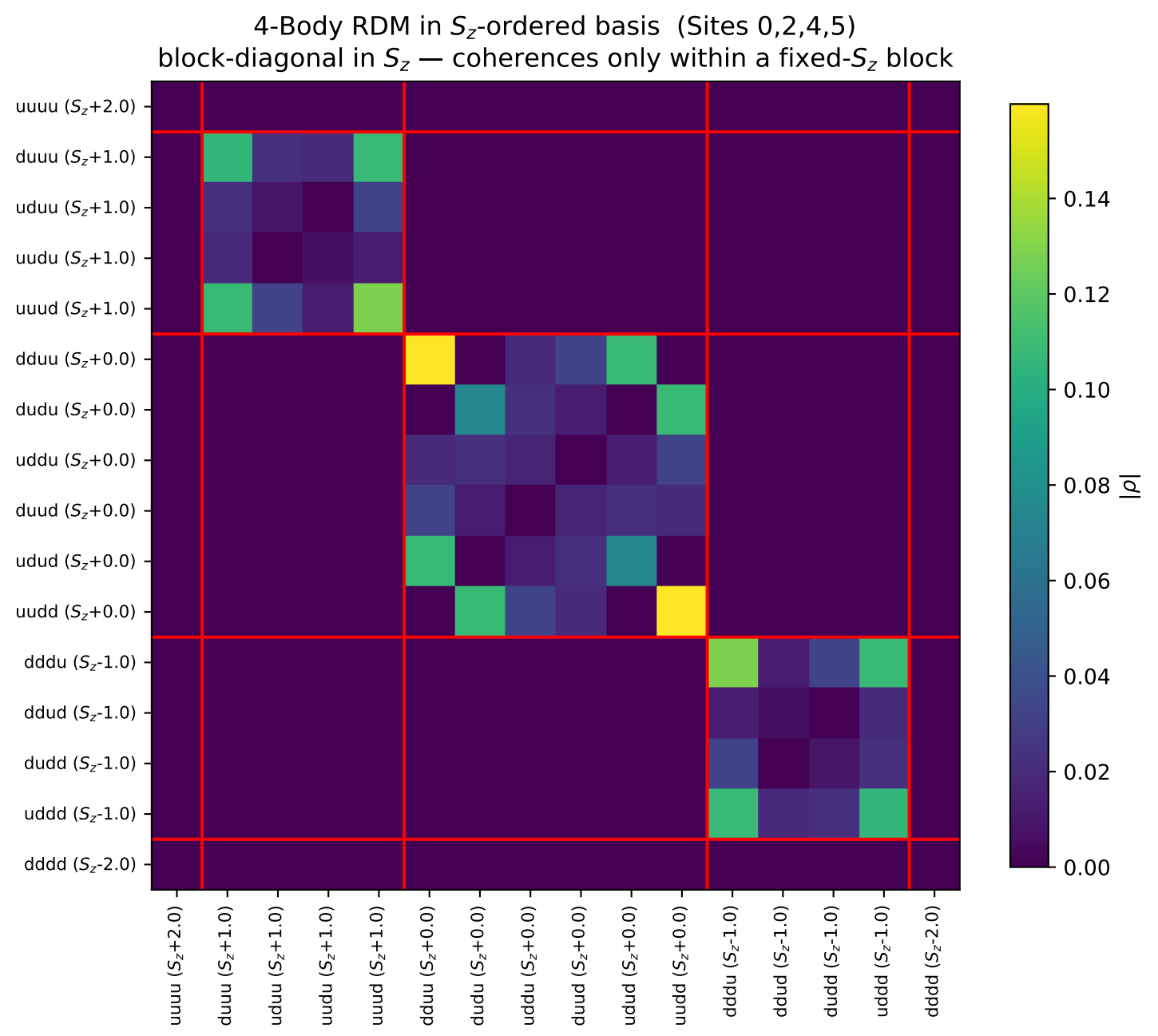
  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = nan   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.37226266

  Status: inconclusive

  Solver source:
    –
```

```
1. p=0.000488  C_GME=0.335794
2. p=0.162017  C_GME=0.002518
3. p=0.161773  C_GME=0.003716
4. p=0.085669  C_GME=0.001211
5. p=0.087991  C_GME=0.004333
... and 5 more states
```


4-Body RDM (Sites 0,2,4,5)
Real Part
|↑↑↑↑⟩ |↑↑↑↓⟩ |↑↑↓↑⟩ |↑↑↓↓⟩ |↑↓↑↑⟩ |↑↓↑↓⟩ |↑↓↓↑⟩ |↑↓↓↓⟩ |↓↑↑↑⟩ |↓↑↑↓⟩ |↓↑↓↑⟩ |↓↑↓↓⟩ |↓↓↑↑⟩ |↓↓↑↓⟩ |↓↓↓↑⟩ |↓↓↓↓⟩
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
Trace=1.000000 Purity=0.228456 Entropy=1.559337 (Imaginary part < 1e-6, not shown)


4-Body RDM in S_z-ordered basis (Sites 0,2,4,5)
block-diagonal in S_z — coherences only within a fixed-S_z block
uuuu (S_z+2.0)
duuu (S_z+1.0)
uduu (S_z+1.0)
uudu (S_z+1.0)
uuud (S_z+1.0)
dduu (S_z+0.0)
dudu (S_z+0.0)
uddu (S_z+0.0)
duud (S_z+0.0)
udud (S_z+0.0)
uudd (S_z+0.0)
dddu (S_z-1.0)
ddud (S_z-1.0)
dudd (S_z-1.0)
uddd (S_z-1.0)
dddd (S_z-2.0)
0.14
0.12
0.10
0.08
0.06
0.04
0.02
0.00
$|\rho|$
Max |coherence between different S_z| = 0.00e+00 ($\Rightarrow S_z$ conserved). Red lines separate S_z sectors.

Analysis Summary - 4-Body RDM (Sites 0,2,4,5)

RDM Metrics
==========

Trace(rho)	1.0000000000
Purity Tr(rho^2)	0.2284561949
Von Neumann entropy	1.5593374628
Eigenvalue min	-4.063173e-17
Eigenvalue max	2.499424e-01
Imaginary part	< 1e-6 (not shown)

GME Analysis
==========

gamma = Tr(rho^2) = 0.22845619

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.

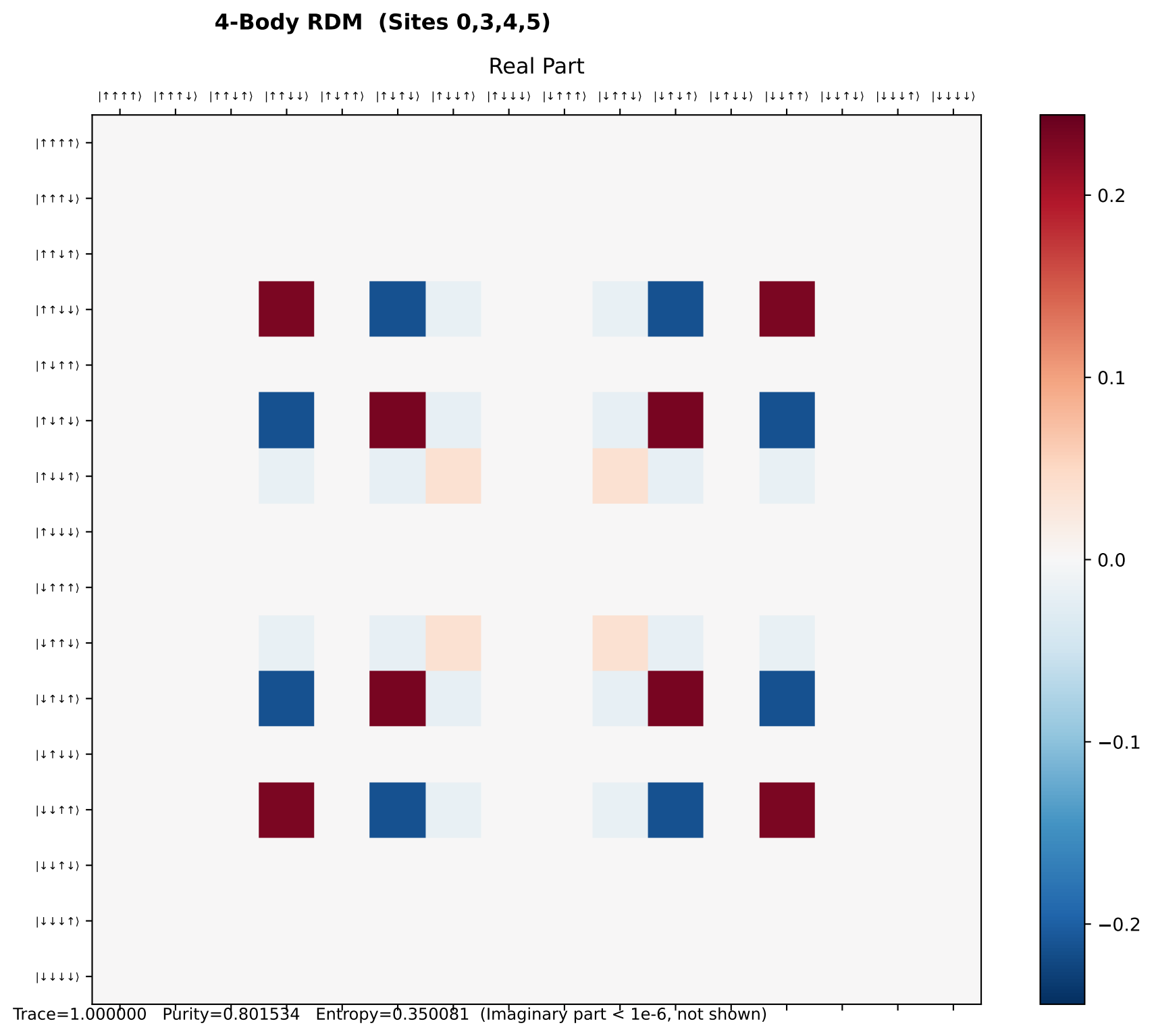
Convex Roof Analysis
==========

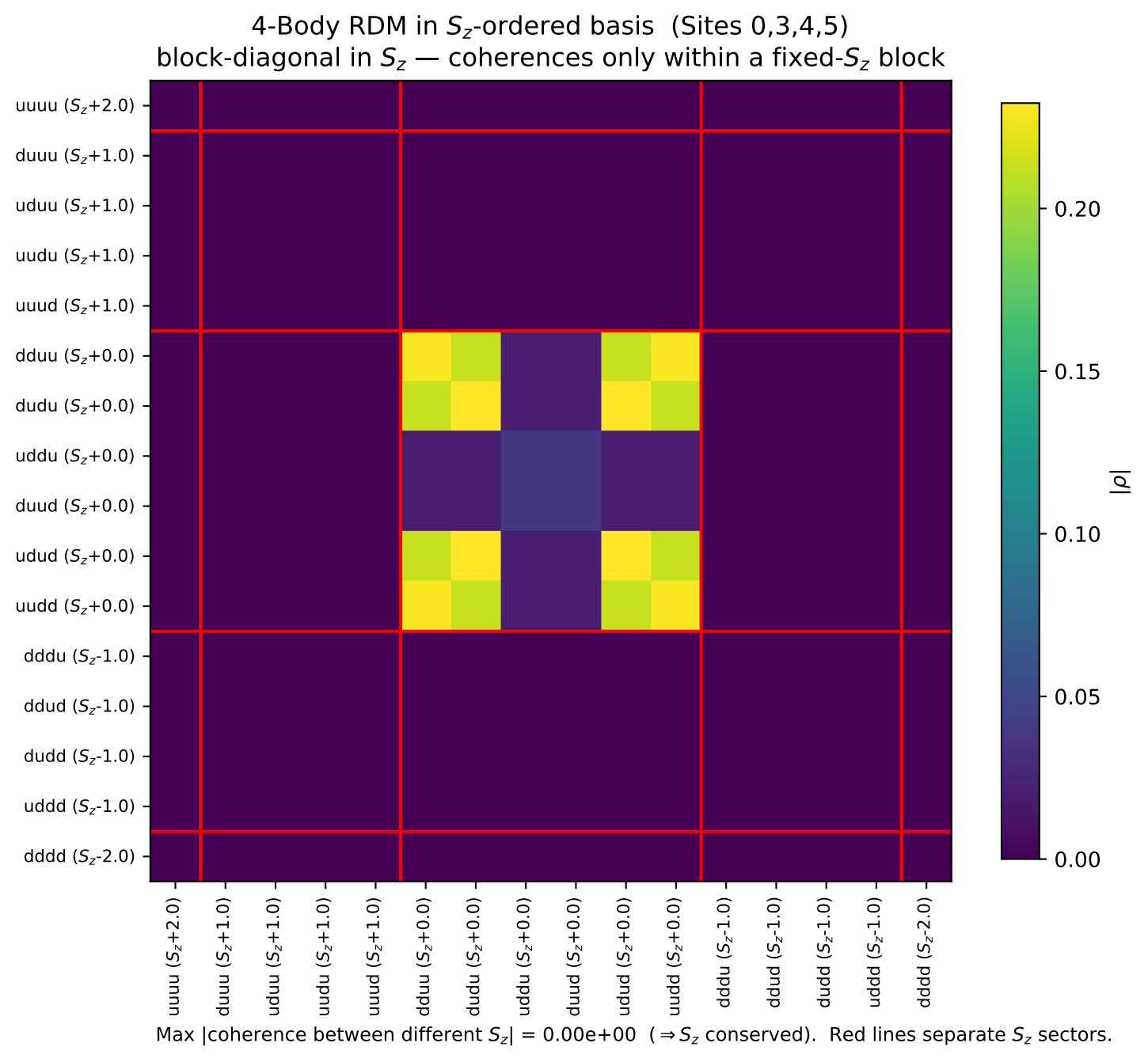
Lower Bounds:
C_LB = 0.00000000
C_GME (unitary optimization):
C_stiefel (unitary opt) = nan BEST
C_GME (unitary optimization - Expoenetial Map):
Upper Bound:
C_UB (eigendecomp) = 0.37293380

Status: inconclusive

Solver source:
—

1. p=0.166679 C_GME=0.001036
2. p=0.167025 C_GME=0.001138
3. p=0.019956 C_GME=0.013605
4. p=0.020516 C_GME=0.007179
5. p=0.166570 C_GME=0.001158
... and 5 more states


4-Body RDM (Sites 0,3,4,5)
Real Part
|↑↑↑↑⟩ |↑↑↑↓⟩ |↑↑↓↑⟩ |↑↑↓↓⟩ |↑↓↑↑⟩ |↑↓↑↓⟩ |↑↓↓↑⟩ |↑↓↓↓⟩ |↓↑↑↑⟩ |↓↑↑↓⟩ |↓↑↓↑⟩ |↓↑↓↓⟩ |↓↓↑↑⟩ |↓↓↑↓⟩ |↓↓↓↑⟩ |↓↓↓↓⟩
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=0.801534 Entropy=0.350081 (Imaginary part < 1e-6, not shown)


4-Body RDM in S_z-ordered basis (Sites 0,3,4,5)
block-diagonal in S_z — coherences only within a fixed-S_z block
uuuu (S_z+2.0)
duuu (S_z+1.0)
uduu (S_z+1.0)
uudu (S_z+1.0)
uuud (S_z+1.0)
dduu (S_z+0.0)
dudu (S_z+0.0)
uddu (S_z+0.0)
duud (S_z+0.0)
udud (S_z+0.0)
uudd (S_z+0.0)
dddu (S_z-1.0)
ddud (S_z-1.0)
dudd (S_z-1.0)
uddd (S_z-1.0)
dddd (S_z-2.0)
0.20
0.15
0.10
0.05
0.00
$|\rho|$
Max |coherence between different S_z| = 0.00e+00 ($\Rightarrow S_z$ conserved). Red lines separate S_z sectors.

Analysis Summary - 4-Body RDM (Sites 0,3,4,5)

RDM Metrics
==========

Trace(rho)	1.0000000000
Purity Tr(rho^2)	0.8015339801
Von Neumann entropy	0.3500813350
Eigenvalue min	-9.828702e-17
Eigenvalue max	8.882873e-01
Imaginary part	< 1e-6 (not shown)

GME Analysis
==========

gamma = Tr(rho^2) = 0.80153398

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.

Convex Roof Analysis
==========

Lower Bounds:
- C_LB = 0.46478952

C_GME (unitary optimization):
- C_stiefel (unitary opt) = nan BEST

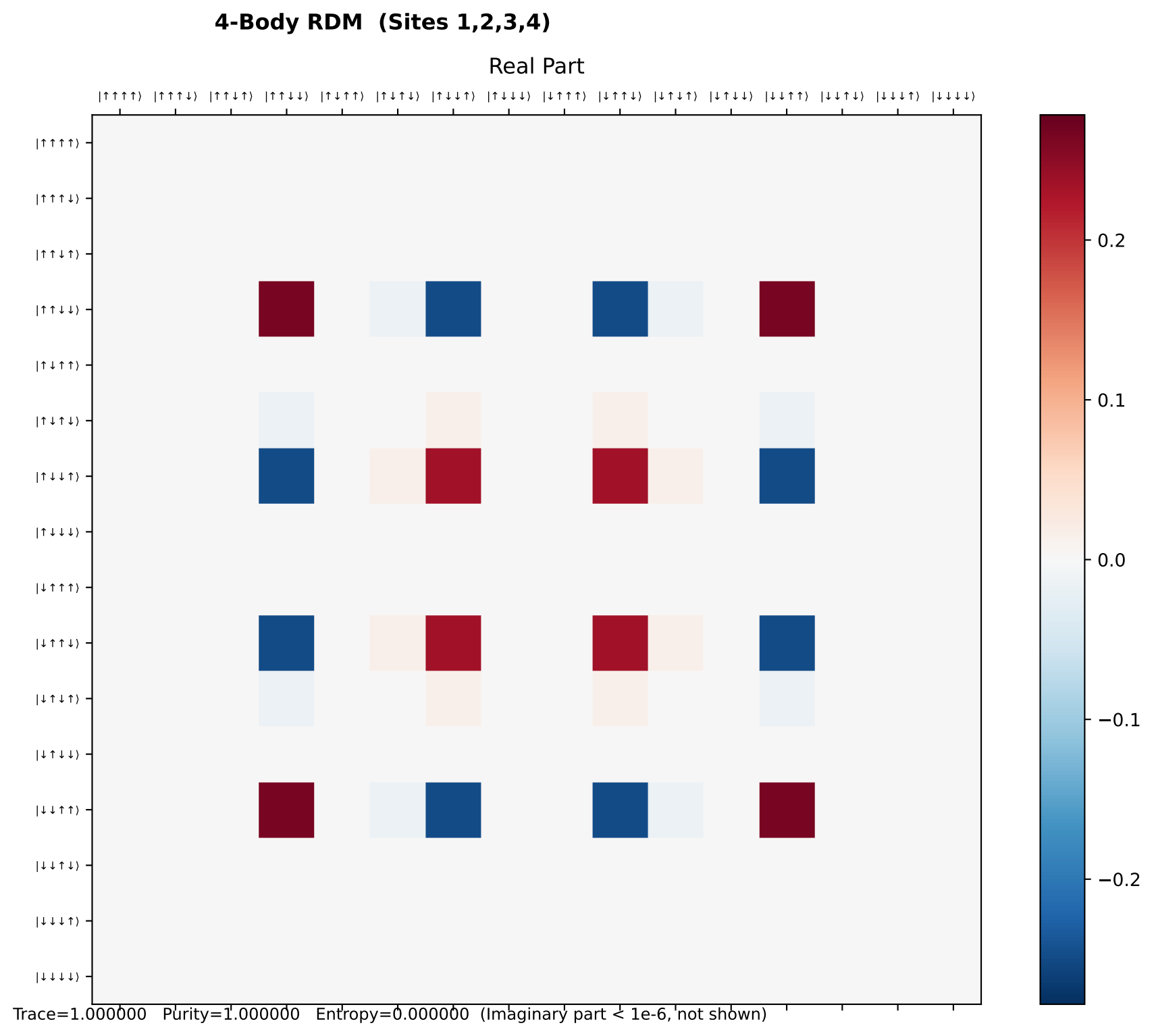
C_GME (unitary optimization - Expoenetial Map):

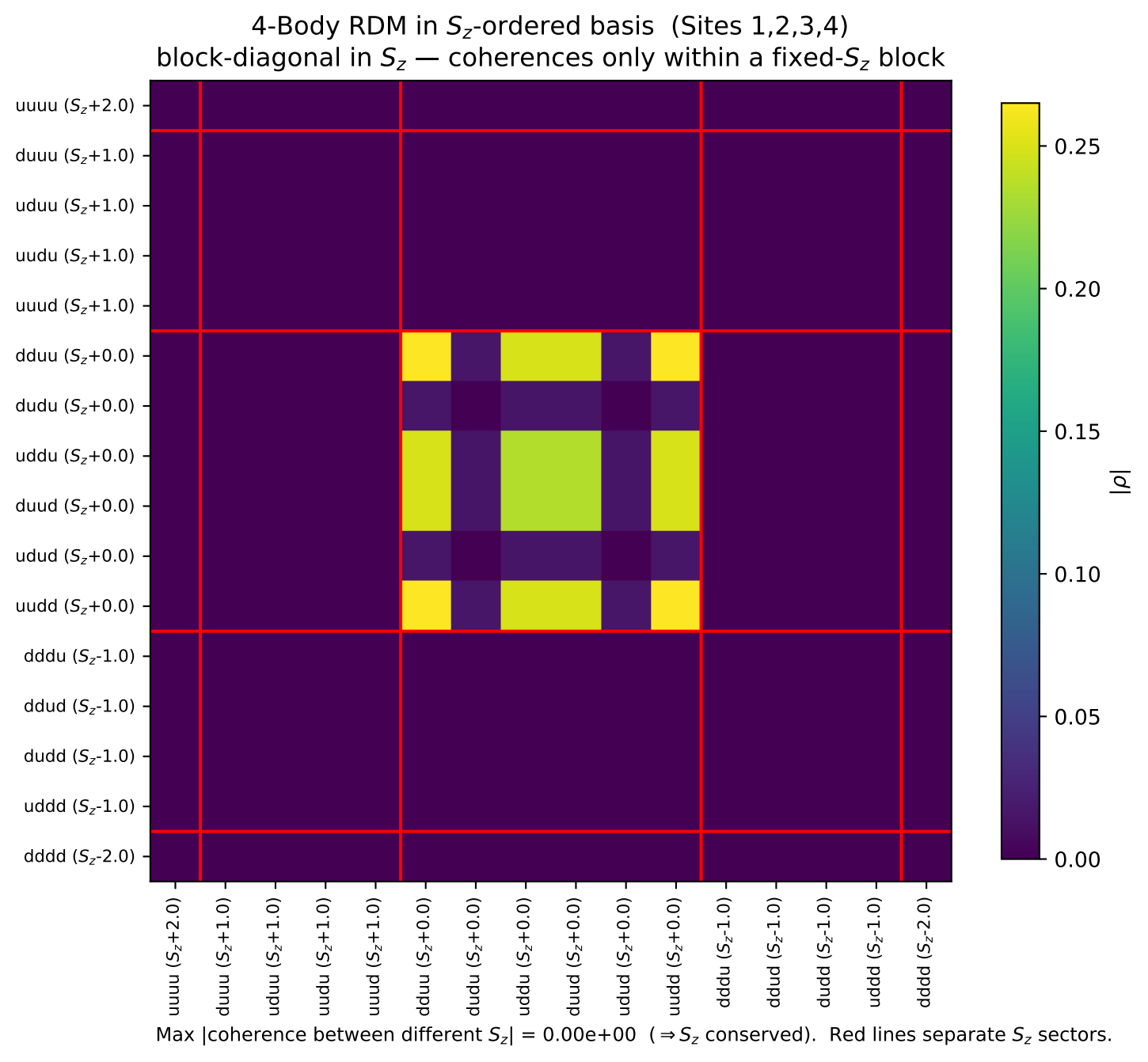
Upper Bound:
- C_UB (eigendecomp) = 0.10938927

Status: no_GME

Solver source:
- –

1. p=0.070421 C_GME=0.000000
2. p=0.078551 C_GME=0.000001
3. p=0.327831 C_GME=0.000000
4. p=0.523198 C_GME=0.000000


4-Body RDM (Sites 1,2,3,4)
Real Part
|↑↑↑↑⟩ |↑↑↑↓⟩ |↑↑↓↑⟩ |↑↑↓↓⟩ |↑↓↑↑⟩ |↑↓↑↓⟩ |↑↓↓↑⟩ |↑↓↓↓⟩ |↓↑↑↑⟩ |↓↑↑↓⟩ |↓↑↓↑⟩ |↓↑↓↓⟩ |↓↓↑↑⟩ |↓↓↑↓⟩ |↓↓↓↑⟩ |↓↓↓↓⟩
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=1.000000 Entropy=0.000000 (Imaginary part < 1e-6, not shown)


4-Body RDM in S_z-ordered basis (Sites 1,2,3,4)
block-diagonal in S_z — coherences only within a fixed-S_z block
uuuu (S_z+2.0)
duuu (S_z+1.0)
uduu (S_z+1.0)
uudu (S_z+1.0)
uuud (S_z+1.0)
dduu (S_z+0.0)
dudu (S_z+0.0)
uddu (S_z+0.0)
duud (S_z+0.0)
udud (S_z+0.0)
uudd (S_z+0.0)
dddu (S_z-1.0)
ddud (S_z-1.0)
dudd (S_z-1.0)
uddd (S_z-1.0)
dddd (S_z-2.0)
0.25
0.20
0.15
0.10
0.05
0.00
$|\rho|$
Max |coherence between different S_z| = 0.00e+00 ($\Rightarrow S_z$ conserved). Red lines separate S_z sectors.

Analysis Summary - 4-Body RDM (Sites 1,2,3,4)

RDM Metrics
==========

Trace(rho)	1.0000000000
Purity Tr(rho^2)	0.9999999998
Von Neumann entropy	0.0000000019
Eigenvalue min	-6.605862e-17
Eigenvalue max	1.000000e+00
Imaginary part	< 1e-6 (not shown)

GME Analysis
==========

gamma = Tr(rho^2) = 1.00000000

A	B	Tr(rho_A^2)
(1,)	(2, 3, 4)	0.50000000
(1, 2)	(3, 4)	0.25271107
(1, 3)	(2, 4)	0.99421839
(1, 4)	(2, 3)	0.25307054
(1, 2, 3)	(4,)	0.50000000
(1, 2, 4)	(3,)	0.50000000
(1, 3, 4)	(2,)	0.50000000

STATUS: GME EXISTS

C_GME = 0.10753238
GME confirmed across ALL cuts.

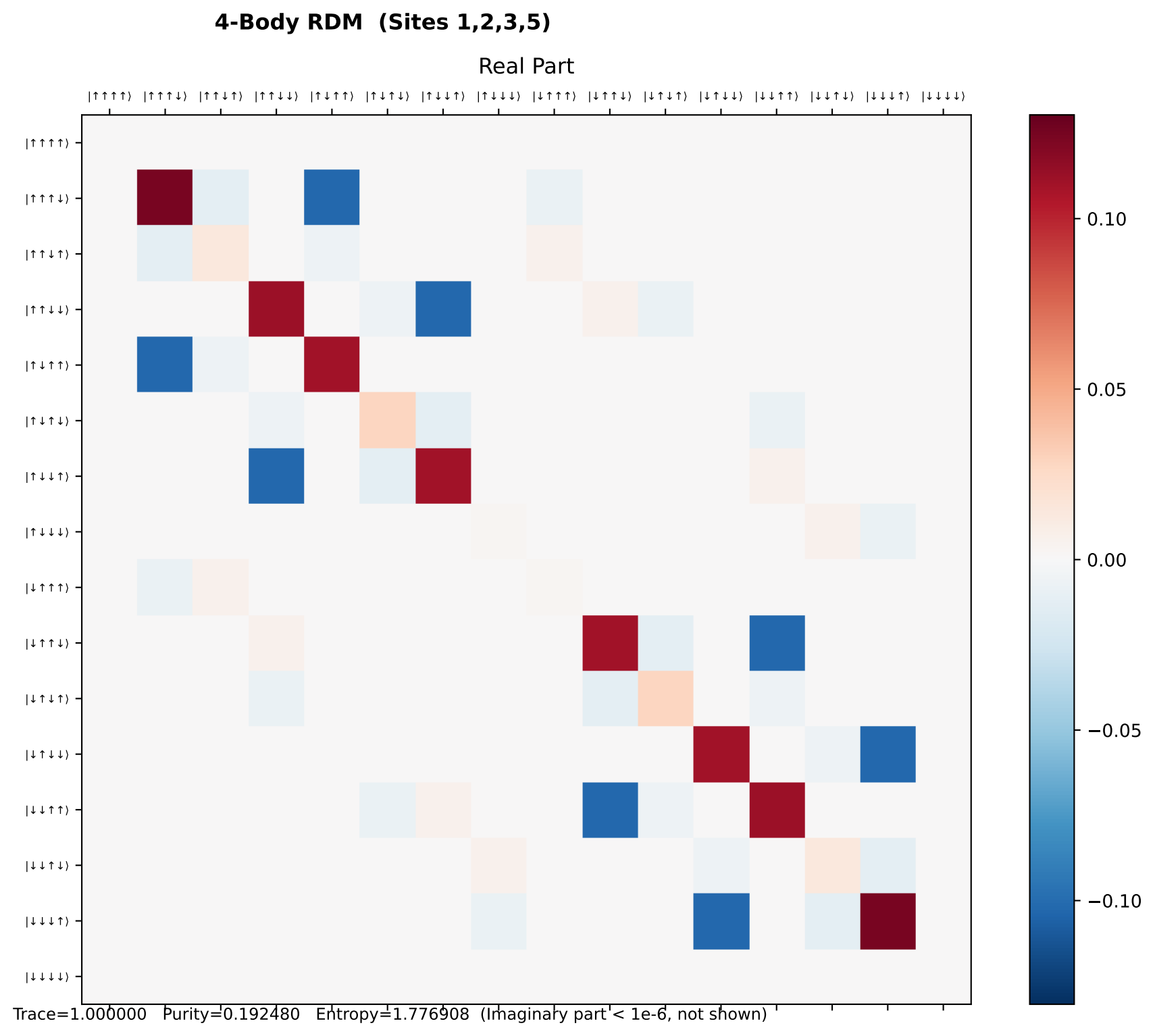
Convex Roof Analysis
==========

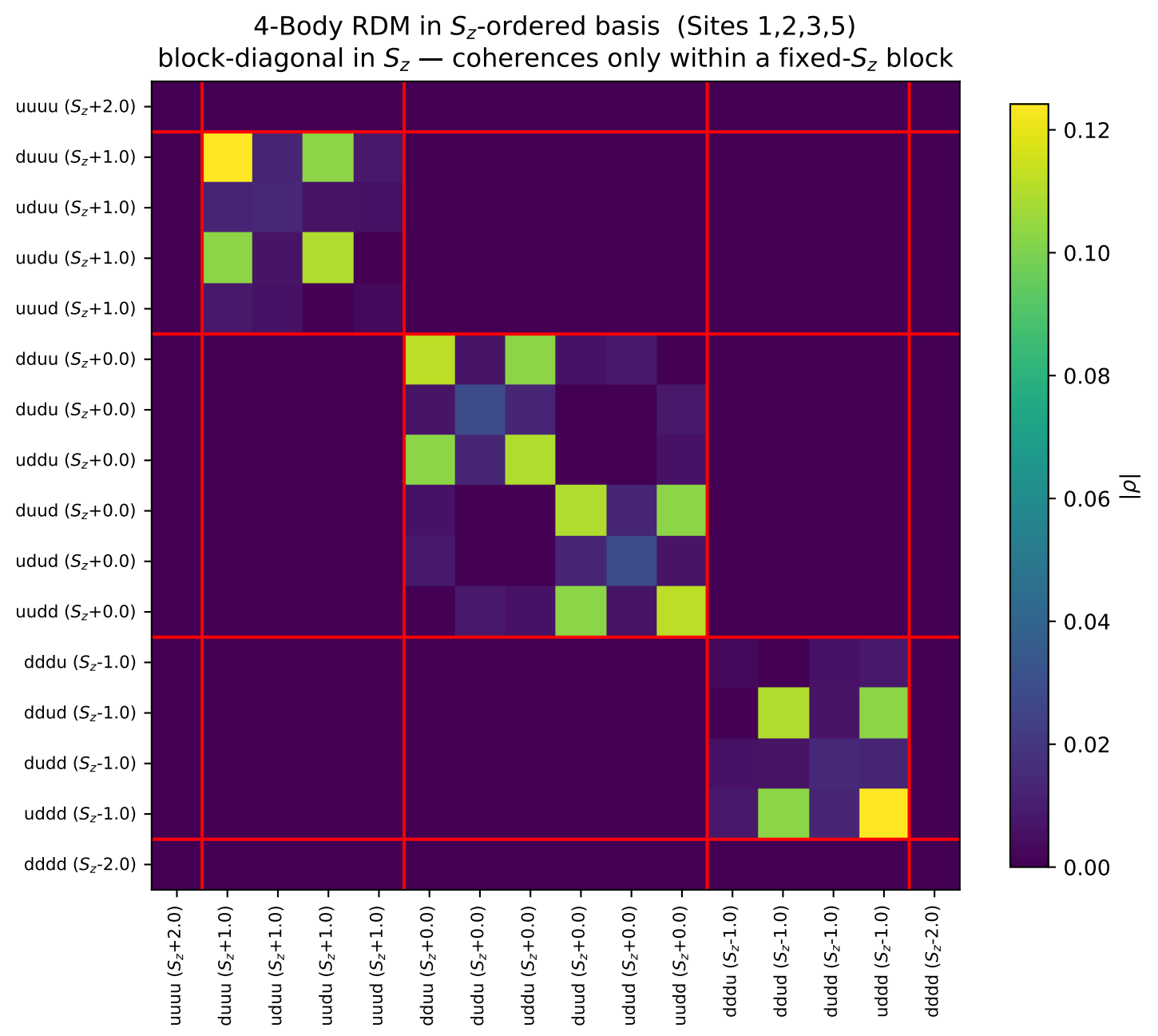
Lower Bounds:
  C_LB = 0.53006144
C_GME (unitary optimization):
  C_stiefel (unitary opt) = nan BEST
C_GME (unitary optimization - Expoenetial Map):
Upper Bound:
  C_UB (eigendecomp) = 0.10753238

Status: GME

Solver source:
  –

1. p=0.898040 C_GME=0.107532
2. p=0.101960 C_GME=0.107532


4-Body RDM (Sites 1,2,3,5)
Real Part
|↑↑↑↑⟩ |↑↑↑↓⟩ |↑↑↓↑⟩ |↑↑↓↓⟩ |↑↓↑↑⟩ |↑↓↑↓⟩ |↑↓↓↑⟩ |↑↓↓↓⟩ |↓↑↑↑⟩ |↓↑↑↓⟩ |↓↑↓↑⟩ |↓↑↓↓⟩ |↓↓↑↑⟩ |↓↓↑↓⟩ |↓↓↓↑⟩ |↓↓↓↓⟩
0.10
0.05
0.00
−0.05
−0.10
Trace=1.000000 Purity=0.192480 Entropy=1.776908 (Imaginary part < 1e-6, not shown)


4-Body RDM in S_z-ordered basis (Sites 1,2,3,5)
block-diagonal in S_z — coherences only within a fixed-S_z block
uuuu (S_z+2.0)
duuu (S_z+1.0)
uduu (S_z+1.0)
uudu (S_z+1.0)
uuud (S_z+1.0)
dduu (S_z+0.0)
dudu (S_z+0.0)
uddu (S_z+0.0)
duud (S_z+0.0)
udud (S_z+0.0)
uudd (S_z+0.0)
dddu (S_z-1.0)
ddud (S_z-1.0)
dudd (S_z-1.0)
uddd (S_z-1.0)
dddd (S_z-2.0)
0.12
0.10
0.08
0.06
0.04
0.02
0.00
$|\rho|$
Max |coherence between different S_z| = 0.00e+00 ($\Rightarrow S_z$ conserved). Red lines separate S_z sectors.

Analysis Summary - 4-Body RDM (Sites 1,2,3,5)

RDM Metrics
==========

Trace(rho)	1.0000000000
Purity Tr(rho^2)	0.1924804145
Von Neumann entropy	1.7769084571
Eigenvalue min	-5.534534e-17
Eigenvalue max	2.198107e-01
Imaginary part	< 1e-6 (not shown)

GME Analysis
==========

gamma = Tr(rho^2) = 0.19248041

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.

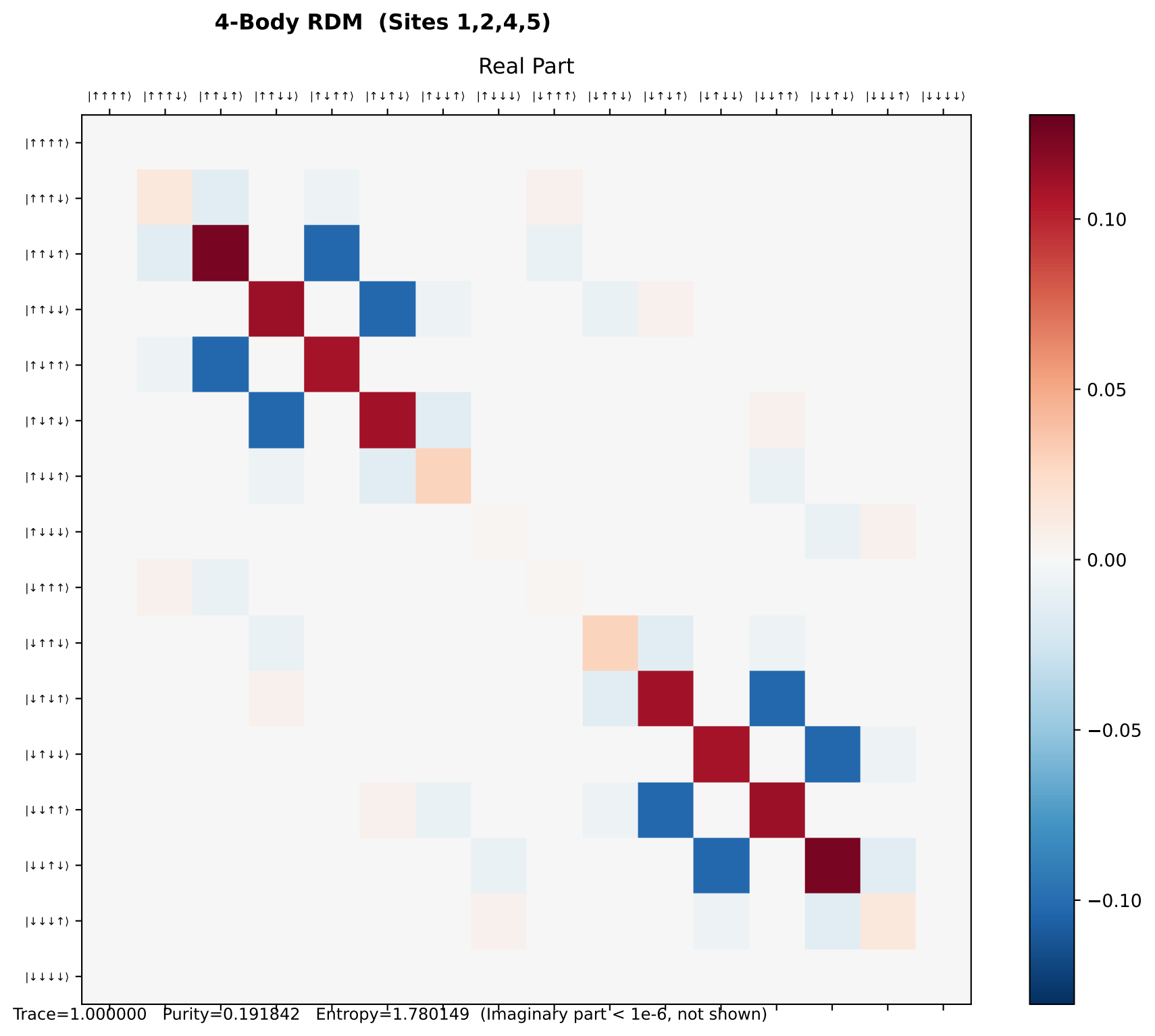
Convex Roof Analysis
==========

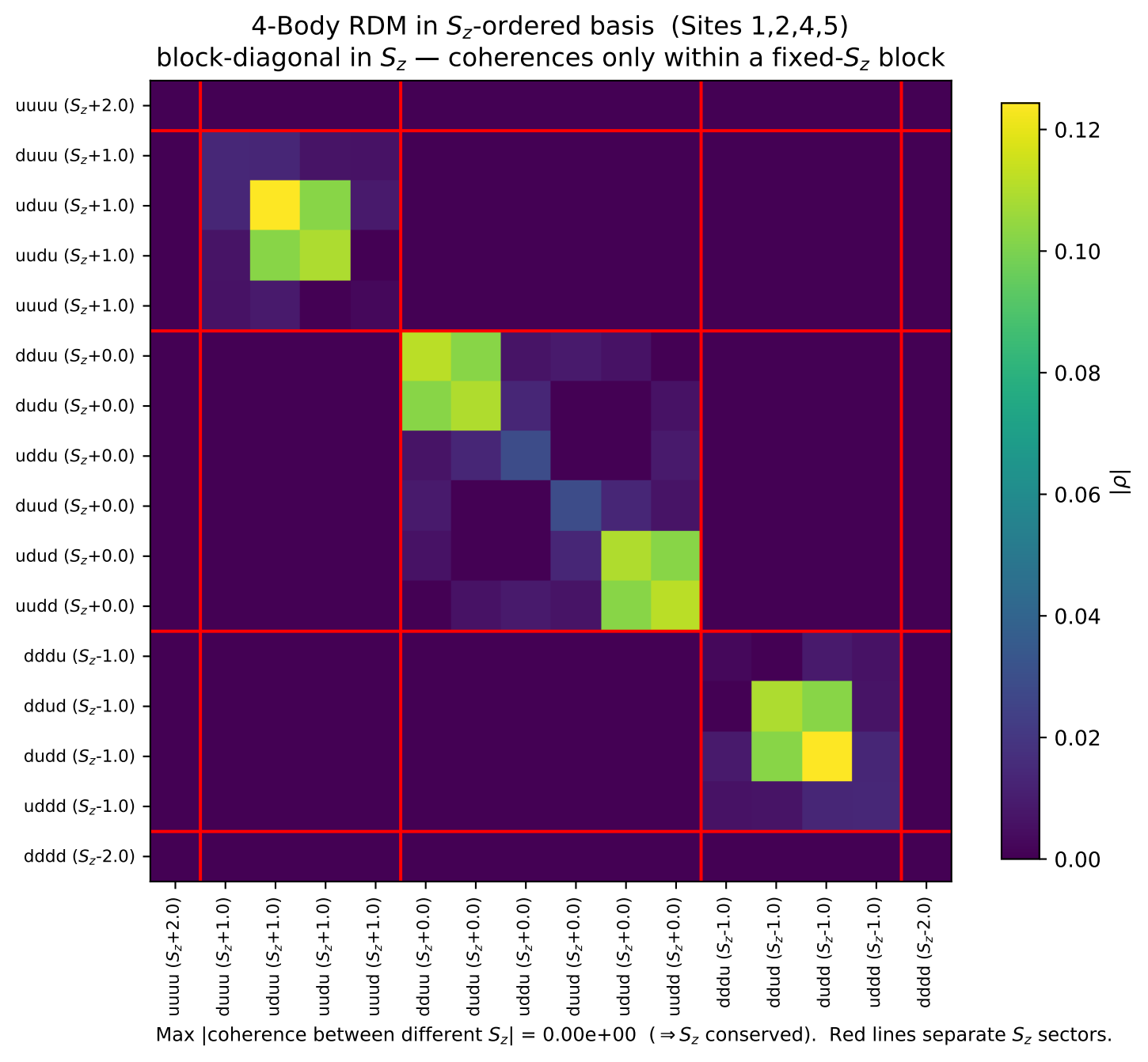
Lower Bounds:
C_LB = 0.00000000
C_GME (unitary optimization):
C_stiefel (unitary opt) = nan BEST
C_GME (unitary optimization - Expoenetial Map):
Upper Bound:
C_UB (eigendecomp) = 0.15050630

Status: inconclusive

Solver source:
–

1. p=0.037861 C_GME=0.097410
2. p=0.204474 C_GME=0.001035
3. p=0.038948 C_GME=0.130090
4. p=0.189096 C_GME=0.079036
5. p=0.157778 C_GME=0.016763
... and 5 more states


4-Body RDM (Sites 1,2,4,5)
Real Part
|↑↑↑↑⟩ |↑↑↑↓⟩ |↑↑↓↑⟩ |↑↑↓↓⟩ |↑↓↑↑⟩ |↑↓↑↓⟩ |↑↓↓↑⟩ |↑↓↓↓⟩ |↓↑↑↑⟩ |↓↑↑↓⟩ |↓↑↓↑⟩ |↓↑↓↓⟩ |↓↓↑↑⟩ |↓↓↑↓⟩ |↓↓↓↑⟩ |↓↓↓↓⟩
0.10
0.05
0.00
−0.05
−0.10
Trace=1.000000 Purity=0.191842 Entropy=1.780149 (Imaginary part < 1e-6, not shown)


4-Body RDM in S_z-ordered basis (Sites 1,2,4,5)
block-diagonal in S_z — coherences only within a fixed-S_z block
uuuu (S_z+2.0)
duuu (S_z+1.0)
uduu (S_z+1.0)
uudu (S_z+1.0)
uuud (S_z+1.0)
dduu (S_z+0.0)
dudu (S_z+0.0)
uddu (S_z+0.0)
duud (S_z+0.0)
udud (S_z+0.0)
uudd (S_z+0.0)
dddu (S_z-1.0)
ddud (S_z-1.0)
dudd (S_z-1.0)
uddd (S_z-1.0)
dddd (S_z-2.0)
0.12
0.10
0.08
0.06
0.04
0.02
0.00
$|\rho|$
Max |coherence between different S_z| = 0.00e+00 ($\Rightarrow S_z$ conserved). Red lines separate S_z sectors.

Analysis Summary - 4-Body RDM (Sites 1,2,4,5)

RDM Metrics
==========

Trace(rho)	1.0000000000
Purity Tr(rho^2)	0.1918423382
Von Neumann entropy	1.7801492269
Eigenvalue min	-1.707988e-17
Eigenvalue max	2.194390e-01
Imaginary part	< 1e-6 (not shown)

GME Analysis
==========

gamma = Tr(rho^2) = 0.19184234

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.

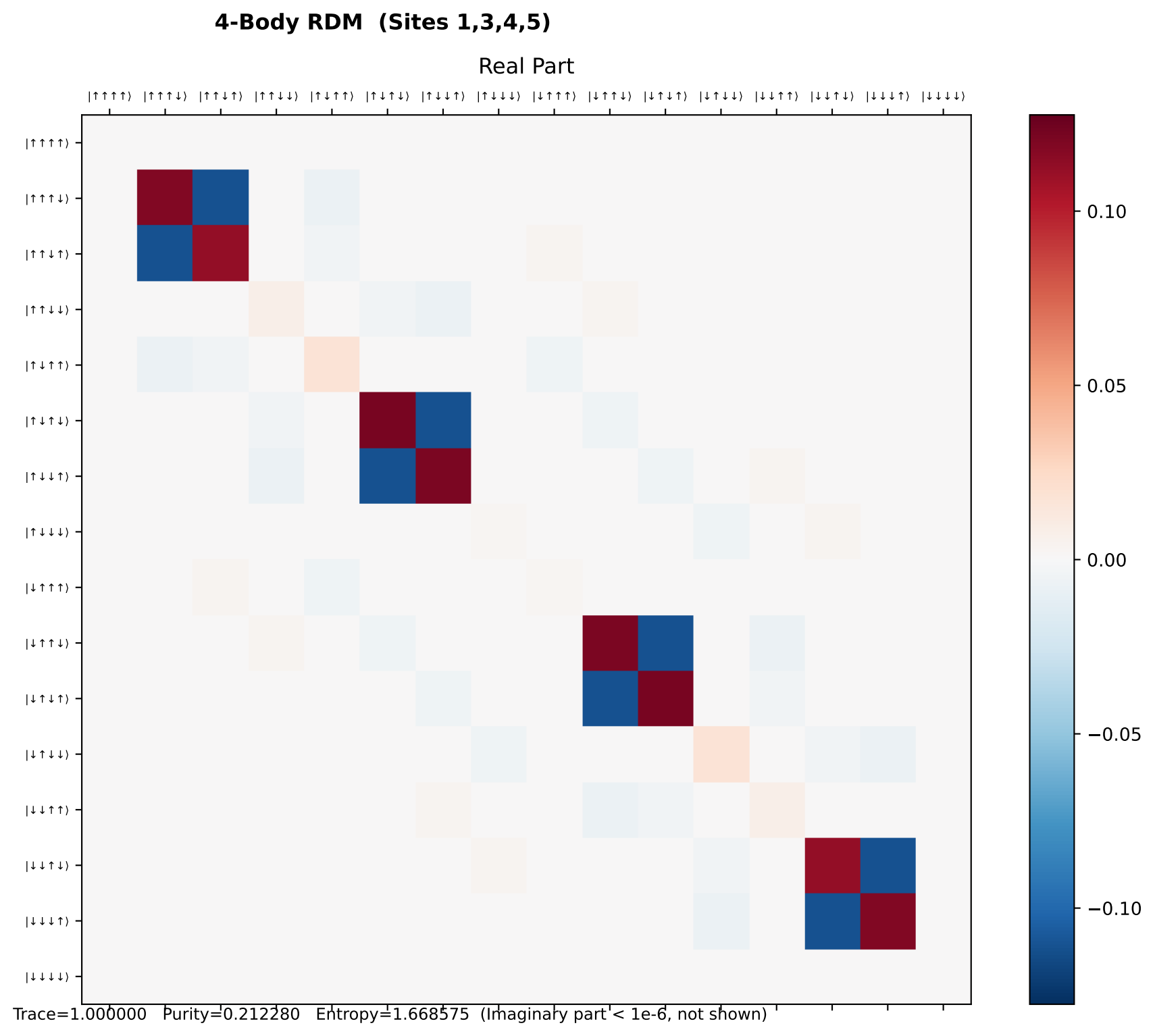
Convex Roof Analysis
==========

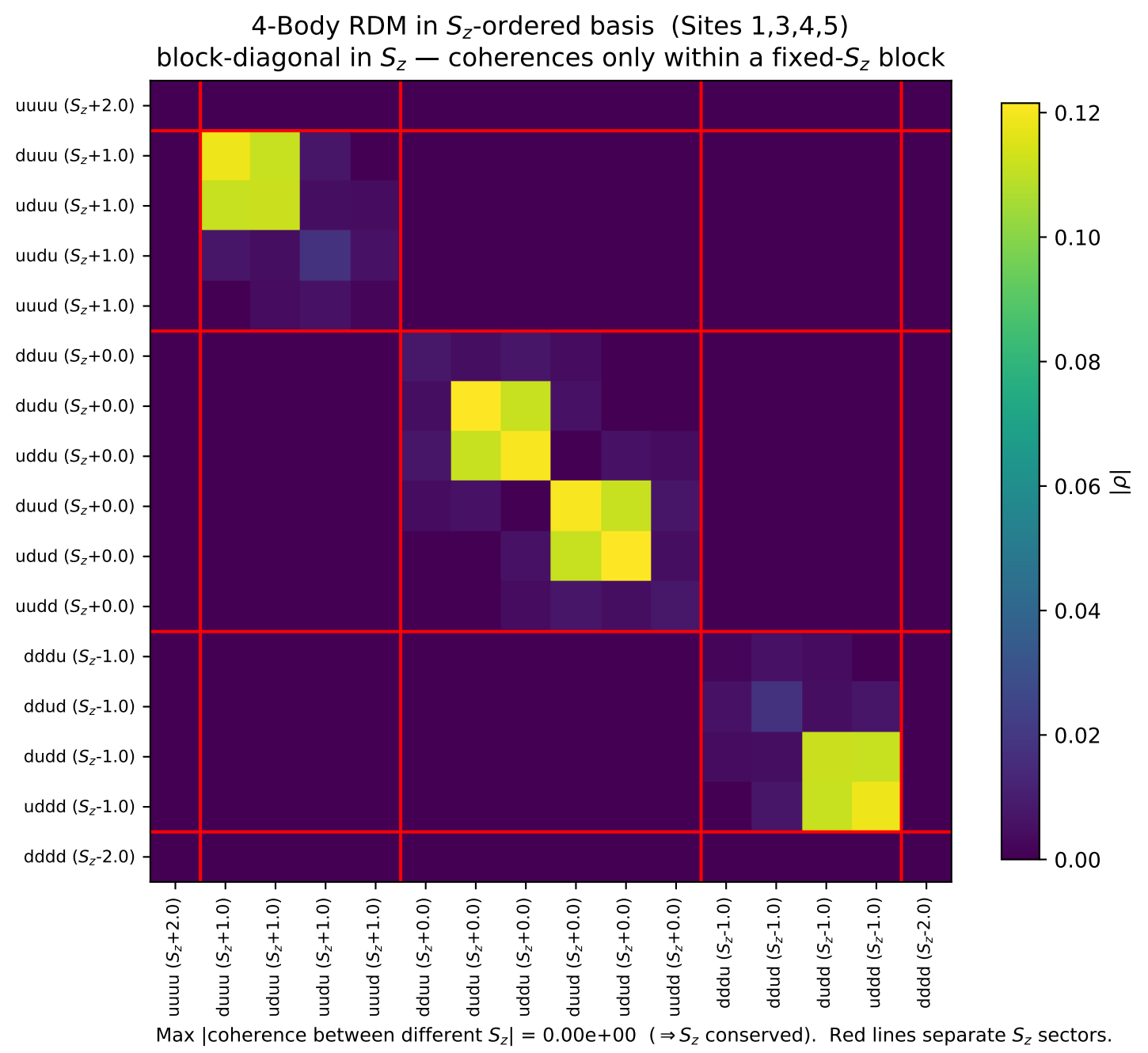
Lower Bounds:
C_LB = 0.00000000
C_GME (unitary optimization):
C_stiefel (unitary opt) = nan BEST
C_GME (unitary optimization - Expoenetial Map):
Upper Bound:
C_UB (eigendecomp) = 0.15383465

Status: inconclusive

Solver source:
–

1. p=0.036726 C_GME=0.249532
2. p=0.208138 C_GME=0.039102
3. p=0.036262 C_GME=0.084932
4. p=0.015953 C_GME=0.096933
5. p=0.195960 C_GME=0.001024
... and 5 more states


4-Body RDM (Sites 1,3,4,5)
Real Part
|↑↑↑↑⟩ |↑↑↑↓⟩ |↑↑↓↑⟩ |↑↑↓↓⟩ |↑↓↑↑⟩ |↑↓↑↓⟩ |↑↓↓↑⟩ |↑↓↓↓⟩ |↓↑↑↑⟩ |↓↑↑↓⟩ |↓↑↓↑⟩ |↓↑↓↓⟩ |↓↓↑↑⟩ |↓↓↑↓⟩ |↓↓↓↑⟩ |↓↓↓↓⟩
0.10
0.05
0.00
−0.05
−0.10
Trace=1.000000 Purity=0.212280 Entropy=1.668575 (Imaginary part < 1e-6, not shown)


4-Body RDM in S_z-ordered basis (Sites 1,3,4,5)
block-diagonal in S_z — coherences only within a fixed-S_z block
uuuu (S_z+2.0)
duuu (S_z+1.0)
uduu (S_z+1.0)
uudu (S_z+1.0)
uuud (S_z+1.0)
dduu (S_z+0.0)
dudu (S_z+0.0)
uddu (S_z+0.0)
duud (S_z+0.0)
udud (S_z+0.0)
uudd (S_z+0.0)
dddu (S_z-1.0)
ddud (S_z-1.0)
dudd (S_z-1.0)
uddd (S_z-1.0)
dddd (S_z-2.0)
0.12
0.10
0.08
0.06
0.04
0.02
0.00
$|\rho|$
Max |coherence between different S_z| = 0.00e+00 ($\Rightarrow S_z$ conserved). Red lines separate S_z sectors.

Analysis Summary - 4-Body RDM (Sites 1,3,4,5)

RDM Metrics
==========

Trace(rho)	1.0000000000
Purity Tr(rho^2)	0.2122800809
Von Neumann entropy	1.6685754151
Eigenvalue min	-1.699805e-17
Eigenvalue max	2.380752e-01
Imaginary part	< 1e-6 (not shown)

GME Analysis
==========

gamma = Tr(rho^2) = 0.21228008

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.

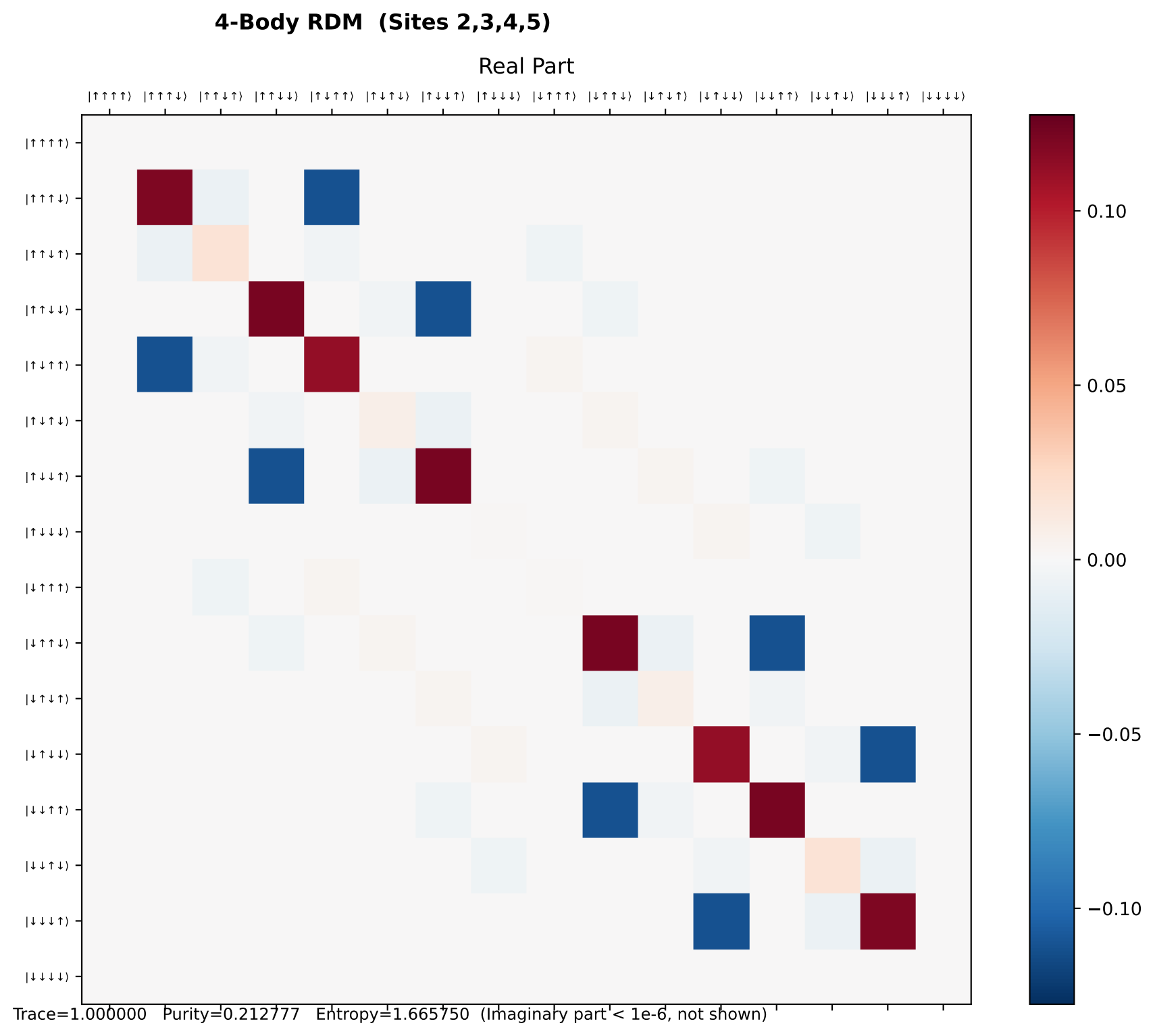
Convex Roof Analysis
==========

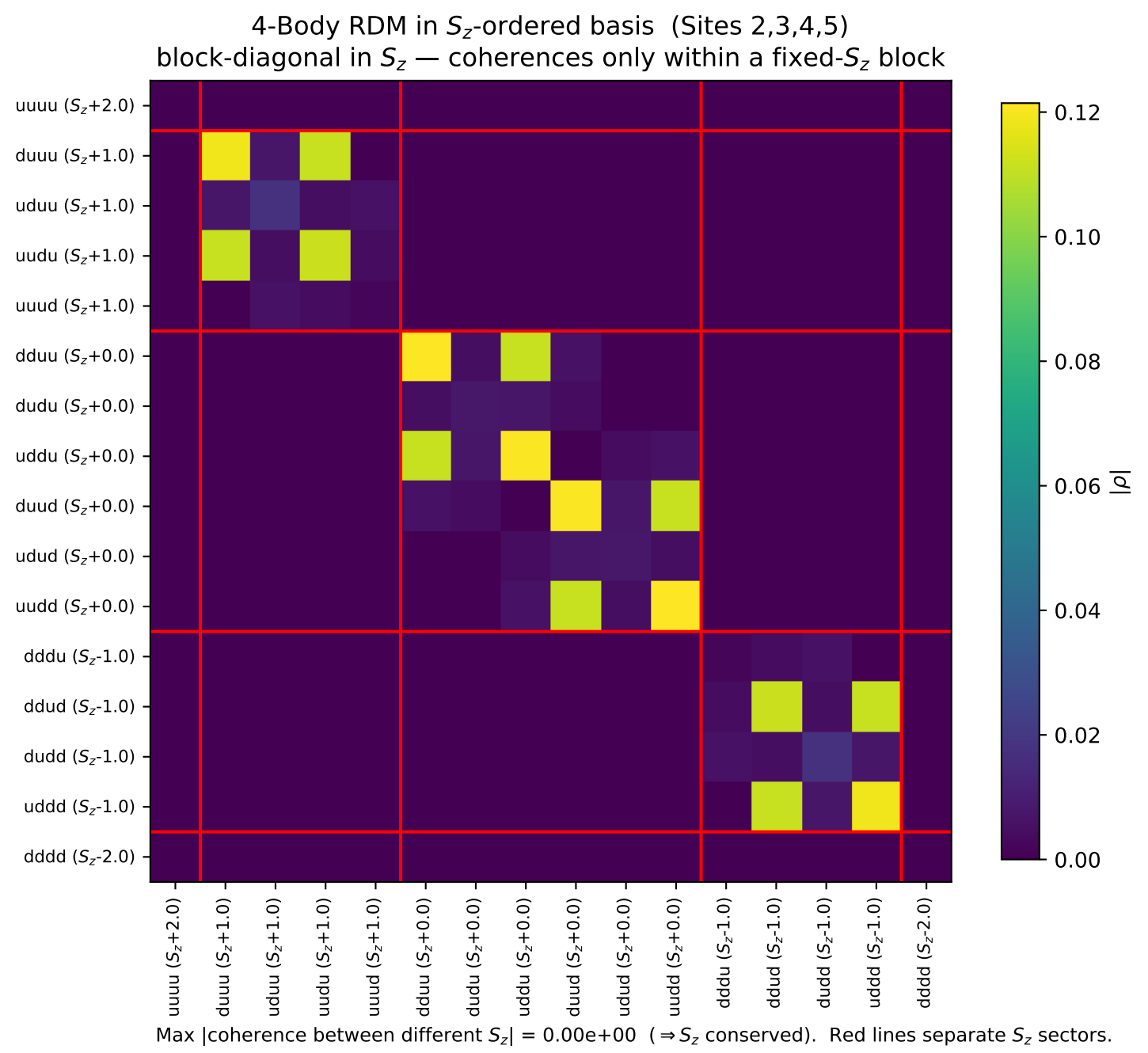
Lower Bounds:
C_LB = 0.00000000
C_GME (unitary optimization):
C_stiefel (unitary opt) = nan BEST
C_GME (unitary optimization - Expoenetial Map):
Upper Bound:
C_UB (eigendecomp) = 0.08037657

Status: inconclusive

Solver source:
–

1. p=0.017160 C_GME=0.381329
2. p=0.114184 C_GME=0.026611
3. p=0.023790 C_GME=0.282247
4. p=0.205909 C_GME=0.000015
5. p=0.167205 C_GME=0.030885
... and 5 more states


4-Body RDM (Sites 2,3,4,5)
Real Part
|↑↑↑↑⟩ |↑↑↑↓⟩ |↑↑↓↑⟩ |↑↑↓↓⟩ |↑↓↑↑⟩ |↑↓↑↓⟩ |↑↓↓↑⟩ |↑↓↓↓⟩ |↓↑↑↑⟩ |↓↑↑↓⟩ |↓↑↓↑⟩ |↓↑↓↓⟩ |↓↓↑↑⟩ |↓↓↑↓⟩ |↓↓↓↑⟩ |↓↓↓↓⟩
0.10
0.05
0.00
−0.05
−0.10
Trace=1.000000 Purity=0.212777 Entropy=1.665750 (Imaginary part < 1e-6, not shown)


4-Body RDM in S_z-ordered basis (Sites 2,3,4,5)
block-diagonal in S_z — coherences only within a fixed-S_z block
uuuu (S_z+2.0)
duuu (S_z+1.0)
uduu (S_z+1.0)
uudu (S_z+1.0)
uuud (S_z+1.0)
dduu (S_z+0.0)
dudu (S_z+0.0)
uddu (S_z+0.0)
duud (S_z+0.0)
udud (S_z+0.0)
uudd (S_z+0.0)
dddu (S_z-1.0)
ddud (S_z-1.0)
dudd (S_z-1.0)
uddd (S_z-1.0)
dddd (S_z-2.0)
0.12
0.10
0.08
0.06
0.04
0.02
0.00
$|\rho|$
Max |coherence between different S_z| = 0.00e+00 ($\Rightarrow S_z$ conserved). Red lines separate S_z sectors.

Analysis Summary - 4-Body RDM (Sites 2,3,4,5)

```
RDM Metrics
===================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.2127765290
  Von Neumann entropy  : 1.6657503887
  Eigenvalue min       : -3.301300e-17
  Eigenvalue max       : 2.381848e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
===================================

  gamma = Tr(rho^2) = 0.21277653

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

STATUS: MIXED STATE
-----------------------------------

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

```
Convex Roof Analysis
===================================

  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = nan   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.07975275

  Status: inconclusive

  Solver source:
    –
```

```
1. p=0.020369  C_GME=0.035552
2. p=0.013784  C_GME=0.000007
3. p=0.020787  C_GME=0.000971
4. p=0.220181  C_GME=0.002730
5. p=0.206069  C_GME=0.038930
... and 5 more states
```